
Optimal Domain-Aware Privacy Mechanisms for Synthetic Data Generation

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Abstract

Differential privacy (DP) imposes fundamental trade-offs between privacy and statistical fidelity in synthetic data generation. While access to public data has been shown to improve these trade-offs empirically, existing approaches use public data only indirectly, through pre-processing (e.g., using pre-trained generative models) or post-processing steps (e.g., matching target statistics estimated from public datasets), while relying on domain-agnostic DP mechanisms. In this work, we lay the theoretical framework to study the principled incorporation of public data into DP mechanisms themselves. We consider normalized histograms as distribution estimators and characterize the asymptotically optimal domain-aware privacy mechanism within a specific class of DP mechanisms. We introduce PUBMIX, a public-data-aware DP mechanism that can be used in histogram-based data synthesis pipelines. Our experiments demonstrate that PUBMIX significantly improves synthetic data generation quality compared to domain-agnostic privacy mechanisms.

1. Introduction

Modern data-driven systems require access to large, diverse datasets in order to train, evaluate, and validate models. In many of these settings, the underlying data consists of sensitive records such as patient histories or financial transactions that cannot be directly shared or reused due to privacy regulations. Differentially private (DP) synthetic data generation offers a solution to this challenge by constructing a synthetic

dataset that preserves the statistical properties of the original private data with formal DP guarantees (Ponomareva et al., 2025). The resulting synthetic dataset can be freely shared and reused in downstream tasks with no privacy leakage.

To improve utility of DP synthetic data generation, several works have explored leveraging auxiliary public data from the same domain, typically incorporating it through pre-processing or post-processing steps. Examples include pre-training generative models on public data (Ghalebikesabi et al., 2023), using public data to inform which statistics to measure (Fuentes et al., 2024; Liu et al., 2021a), and post-processing synthetic data to align with public data constraints (Slavkovic & Reimherr, 2020; Wang et al., 2023). In all these approaches, however, the underlying DP mechanism (how randomness is injected to ensure privacy) remains domain-agnostic, typically based on Gaussian or Laplace noise (Dwork et al., 2006).

We propose a fundamentally different paradigm for incorporating public data: using it directly to design how randomness is injected to achieve privacy in data synthesis. By constructing domain-aware noise distributions informed by public data, we obtain DP mechanisms that achieve improved privacy–utility trade-offs compared to domain-agnostic privacy mechanisms.

DP synthetic data generation typically involves estimating the distribution of a private dataset, perturbing the estimate to ensure privacy, and sampling synthetic data from the perturbed distribution (McKenna et al., 2021; Frigerio et al., 2019; Pfitzner & Arnrich, 2022; McKenna et al., 2022; 2019; Hardt et al., 2012; Zhang et al., 2017). Standard domain-agnostic DP mechanisms, such as Laplace or Gaussian noise, perturb the estimated private distribution in arbitrary directions on the probability simplex. In contrast, we show that public data can be leveraged to identify optimal directions along which the private distribution should be perturbed.

In this paper, we develop a theoretical framework for analyzing a linear mixing-based class of domain-aware DP mechanisms for private data synthesis, considering normalized histograms as private distribution estimators. In this canonical setting, we characterize the asymptotically optimal noise distribution, demonstrating that it can be approximated by a *floor-raised* (Gallager, 1968) version of the public distribution. From a theoretical stand point, these insights serve

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as a foundational step toward developing domain-aware DP mechanisms for complex distribution estimators used in synthetic data generation (see Sec. 3.2). From a practical perspective, our analysis directly informs the design of PUBMIX, a domain-aware DP mechanism for histogram sampling that can replace standard Gaussian or Laplace mechanisms in existing histogram-based synthesis pipelines (Lin et al., 2024; Xie et al., 2024; Abacha et al., 2025; Zhang et al., 2025) for improved utility (see Sec. 5.2). Our key contributions are:

- We introduce a principled framework for leveraging public data to construct domain-aware noise distributions that improve the privacy-utility trade-off in DP data synthesis compared to domain-agnostic privacy mechanisms.
- We provide an information-theoretic analysis of the privacy-utility trade-off in DP histogram sampling within a class of mechanisms and characterize the asymptotically optimal domain-aware privacy mechanism.
- We propose PUBMIX, a DP histogram sampling mechanism that incorporates domain-aware noise and serves as a drop-in privacy module for histogram-sampling-based synthetic data generation.

1.1. Related Work

DP synthetic data generation has been studied across modalities including tabular data, text, and images. Existing approaches include (i) statistical methods that privately release noisy summary statistics to fit distributional models (McKenna et al., 2022; Hardt et al., 2012; Cai et al., 2021; Zhang et al., 2017; McKenna et al., 2021), and (ii) training deep generative models with DP-SGD (Frigerio et al., 2019; Dockhorn et al., 2023; Yue et al., 2023). Complementary paradigms include PATE (Papernot et al., 2018; Long et al., 2021), which trains teachers on disjoint private partitions and aggregates their noisy predictions to label public data for synthesis; Private Evolution (Lin et al., 2024; Xie et al., 2024; Tran et al., 2026), which iteratively samples from (hierarchical) histograms to produce synthetic datasets; and variants of private prediction (Amin et al., 2024; Tang et al., 2024), which synthesize text by privately aggregating next-token predictions from LLMs prompted with sensitive examples. See (Ponomareva et al., 2025) for a comprehensive survey. While some methods leverage public data to reduce privacy cost (Fuentes et al., 2024; Liu et al., 2021a), the underlying DP mechanisms remain largely domain-agnostic (Gaussian, Laplace, or exponential mechanisms). In contrast, we introduce the concept of domain-aware privacy mechanisms.

DP multi-sampling provides sample-complexity bounds for drawing independent samples from a given private distribution under DP, covering broader distribution families

(Cheu & Nayak, 2025). We extend the DP multi-sampling setting by incorporating public data into the problem formulation, while restricting to a more structured setting (linear mixing-based DP mechanisms) to make public-data integration tractable.

Existing linear mixing methods include variants of private prediction (Flemings et al., 2024; Ginart et al., 2022) that serve as alternatives to existing DP training methods. Instead of privatizing the model itself, these methods enforce privacy at inference time by perturbing the model’s output. In LLMs, this typically involves mixing or projecting the private model’s next-token distribution toward a public distribution to limit leakage from memorized private data. While both PUBMIX and private prediction involve linear mixing and public distributions, they differ fundamentally in goal and technical design (More details in App. A).

DP histograms are a core primitive underlying many synthesis pipelines, with work spanning basic privatization (Dwork & Roth, 2014; Dwork et al., 2006), hierarchical constructions (Qardaji et al., 2013), and adaptive/improved binning (Hay et al., 2010). Recent private evolution methods and their federated learning extensions (Abacha et al., 2025; Hou et al., 2025) further highlight the practical role of histogram sampling, demonstrating strong empirical performance across modalities when combined with foundation models. PUBMIX can be incorporated as a drop-in privacy mechanism within the histogram-based pipelines.

2. Problem Formulation

Let $D_{prv} = \{x_1, \dots, x_n\} \subseteq \mathcal{X}^n$ denote a private dataset where $\mathcal{X}$ is a discrete sample space with $|\mathcal{X}| = d$. Each $x_i \in D_{prv}$ is sampled i.i.d. from an underlying distribution $P_{prv} \in \Delta_d$, where Δ_d is the probability simplex over $\mathcal{X}$. Let $D_{pub} = \{v_i\}_{i=1}^\ell \subseteq \mathcal{X}^\ell$ be a public dataset from the same domain as D_{prv} , sampled i.i.d. from an underlying distribution $P_{pub} \in \Delta_d$. We estimate P_{prv} and P_{pub} using the normalized histograms of D_{prv} and D_{pub} over $\mathcal{X}$, denoted by $\bar{P}_{prv}$ and $\bar{P}_{pub}$, respectively.

Our goal is to generate a synthetic dataset $D_{syn} = \{y_i\}_{i=1}^m$ consisting of m samples that: (i) satisfies DP with respect to D_{prv} , (ii) is statistically *close* to D_{prv} . For this, we design a synthetic data distribution $P_{syn} \in \Delta_d$, from which we draw m samples i.i.d. to obtain the synthetic dataset D_{syn} . Specifically, we consider synthetic distributions of the form:

$$P_{syn} = (1 - \beta)\bar{P}_{prv} + \beta P_{nse} \quad (1)$$

where $\beta \in [0, 1]$ is a mixing coefficient and $P_{nse} \in \Delta_d$ is a noise distribution. β and P_{nse} are globally known fixed parameters that are independent of the private data D_{prv} .

Our goal is to find the optimum parameters β and P_{nse} in (1) that achieves the maximum *utility* while satisfying DP.

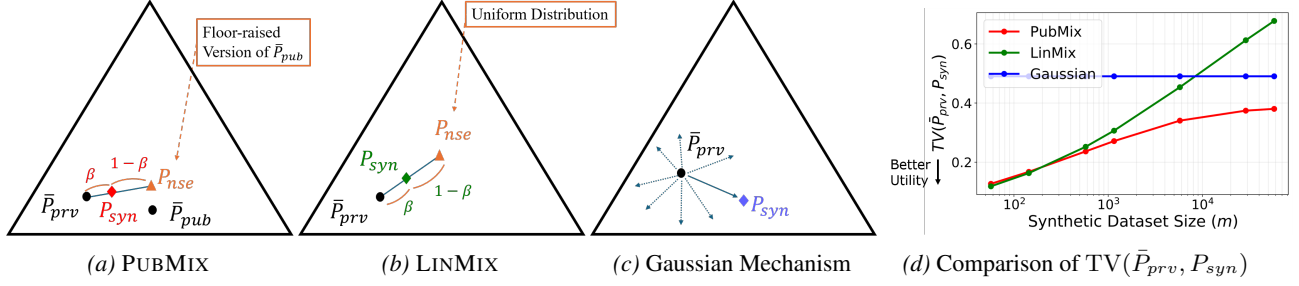


Figure 1. Comparison of PUBMIX, LINMIX, and the Gaussian Mechanism for differentially private histograms. (a)–(c) provide a geometric illustration of the perturbations in the probability simplex for (a) PUBMIX, (b) LINMIX, and (c) the Gaussian mechanism. (d) shows the total variation distance between $(\bar{P}_{prv}, P_{syn})$ as a function of the number of synthetic samples m , for PUBMIX, LINMIX, and the Gaussian mechanism for $n = 57717$, $\varepsilon = 5$, $\delta = 1/n$ based on the 2023 ACS PUMS dataset.

When the mixing parameter $\beta = 0$, we have $P_{syn} = \bar{P}_{prv}$, which is ideal for utility but provides no privacy guarantee. The other extreme, $\beta = 1$, yields $P_{syn} = P_{nse}$, and the generated synthetic data are independent of the private dataset, resulting in a substantial loss of utility while ensuring perfect privacy. Accordingly, for a fixed noise distribution P_{nse} we first identify the smallest β that satisfies DP, and then optimize P_{nse} to obtain the best achievable privacy–utility trade-off. The resulting P_{nse} determines the optimal direction along which the private distribution $\bar{P}_{prv}$ should be perturbed. Here, P_{nse} will depend on the public data, as we will see shortly. First, we formally define the privacy constraint and the utility metric.

Privacy constraint: We consider (ε, δ) -DP of the m generated synthetic samples D_{syn} . Let D_{prv} and D'_{prv} be any two neighboring datasets that differ in only one data entry. Let P_{syn} and P'_{syn} be the corresponding synthetic distributions from (1) for any fixed $\beta \in [0, 1]$ and $P_{nse} \in \Delta_{\mathcal{X}}$. Then, the (ε, δ) -DP constraint is given by,

$$\mathbb{P}(D_{syn} \in S) \leq e^\varepsilon \mathbb{P}(D'_{syn} \in S) + \delta, \quad \forall S \subseteq \mathcal{X}^m, \forall D_{prv}, D'_{prv} \quad (2)$$

where D_{syn} and D'_{syn} correspond to the m i.i.d. samples drawn from P_{syn} and P'_{syn} .

Utility: The utility of D_{syn} is defined as the total variation (TV) distance between $\bar{P}_{prv}$ and P_{syn} , i.e., $\text{TV}(\bar{P}_{prv}, P_{syn})$.

The admissible set: To select a noise distribution P_{nse} in (1), we define an *admissible set* of private distributions *close* to the public distribution $\bar{P}_{pub}$. This set is fixed and independent of the given private data D_{prv} . We define the admissible set as all distributions within a γ -TV ball around $\bar{P}_{pub}$:

$$\mathcal{B}_{\gamma, \bar{P}_{pub}} = \{P \in \Delta_d : \text{TV}(P, \bar{P}_{pub}) \leq \gamma\}$$

for some proximity parameter $\gamma \in (0, 1]$, independent of

D_{prv} . The parameter γ is chosen based on public knowledge (see Sec. 5). The key motivation for this definition is that datasets drawn from the same domain induce distributions that are close to each other on the probability simplex with high probability. We therefore seek a P_{nse} that are optimized for distributions in $\mathcal{B}_{\gamma, \bar{P}_{pub}}$. Concretely, as we describe next, we optimize P_{nse} considering the worst-case private distribution within $\mathcal{B}_{\gamma, \bar{P}_{pub}}$.¹

An optimization formulation for mechanism design:

Given a target (ε, δ) , a public distribution $\bar{P}_{pub}$, and a proximity parameter γ , our goal is to find the optimal noise distribution P_{nse} and mixing coefficient β in (1) that satisfies the privacy constraint in (2) while maximizing utility. The optimization problem we solve to obtain these parameters is given by:

$$\begin{aligned} \min_{P_{nse} \in \Delta_d} \quad & \min_{\beta \in [0, 1]} \quad \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \\ \text{s.t.} \quad & \mathbb{P}(D_{syn} \in S) \leq e^\varepsilon \mathbb{P}(D'_{syn} \in S) + \delta, \\ & \forall S \subseteq \mathcal{X}^m, \forall D_{prv}, D'_{prv} \end{aligned} \quad (3)$$

We denote the parameters that solve (3) as (β^*, P_{nse}^*) .²

PUBMIX: Once the parameters β^* and P_{nse}^* are obtained, for any given D_{prv} , we compute $P_{syn} = (1 - \beta^*)\bar{P}_{prv} + \beta^*P_{nse}^*$ using the corresponding $\bar{P}_{prv}$, and draw m samples from P_{syn} as private synthetic data D_{syn} . We call this mechanism PUBMIX, and it is outlined in Alg. 1.

Why use PUBMIX instead of just adding noise to histograms? Standard DP histogram mechanisms (Dwork & Roth, 2014) add i.i.d. Gaussian or Laplace noise to

¹If the private distribution of D_{prv} lies outside $\mathcal{B}_{\gamma, \bar{P}_{pub}}$ (which cannot be verified a priori), PUBMIX still satisfies DP. However, a weaker utility guarantee will be achieved compared to (3).

²The parameters β^* and P_{nse}^* in PUBMIX are not derived from the given private dataset. They are pre-computed parameters based on the worst case private distribution within a γ -TV ball around $\bar{P}_{pub}$ (see (3)), and are independent of the given private dataset.

each histogram bin to achieve DP. Since the noise is zero-mean and independent across bins, each bin can increase or decrease arbitrarily. When normalized to obtain a probability distribution, the perturbed histogram can move in arbitrary directions on the probability simplex relative to the non-private distribution. This domain-agnostic approach does not consider which perturbation directions optimize the privacy-utility trade-off for the given data domain (Fig. 1c). In contrast, PUBMIX leverages public data from the same domain to identify a principled direction in which to perturb $\bar{P}_{prv}$, yielding a synthetic distribution P_{syn} that better preserves utility while still satisfying DP (Fig. 1a).

A key feature of PUBMIX is *sample dependency*. Most existing DP synthesis methods incur a high privacy cost by releasing the entire privacy-protected distribution P_{syn} (Xie et al., 2024; Lin et al., 2024). For instance, in existing DP histogram-based synthesis methods that add Gaussian or Laplace noise to each of the d bins independently, $TV(\bar{P}_{prv}, P_{syn})$ increases linearly with d (Qardaji et al., 2013). However, many applications require only a finite number of synthetic samples, and not the full distribution.

We show that, when the goal is to release only m synthetic samples from the histogram rather than the histogram itself, it is possible to construct a synthetic sampling distribution P_{syn} that is significantly closer to $\bar{P}_{prv}$, resulting in more accurate synthetic data (Fig. 1-d). The key insight is that the normalized histogram of a private dataset D_{prv} differs from that of any neighboring dataset D'_{prv} in at most two coordinates. Specifically, as n denotes the number of samples in D_{prv} , replacing a single sample causes one bin to increase by $1/n$ while another bin decreases by $1/n$. When releasing an entire DP histogram, this sparsity of the neighboring difference cannot be leveraged, and noise must be injected across all d coordinates. However, when releasing only samples, and working under (ϵ, δ) -DP, this sparse structure can be leveraged, yielding tighter privacy accounting.

3. Main Results: Overview and Intuition

Before presenting the full mathematical treatment of how to solve (3) in Section 4, we first provide a high-level intuition for our solution and how PUBMIX leverages public data.

3.1. Technical overview

Recall that PUBMIX constructs the synthetic dataset D_{syn} by drawing m samples from a carefully designed distribution $P_{syn} = (1 - \beta^*)\bar{P}_{prv} + \beta^*P_{nse}^*$. Here, β^* and P_{nse}^* are the optimal parameters. To solve the optimization in (3), we first fix a noise distribution P_{nse} and find the minimum mixing weight β_{min} (for this specific P_{nse}) for which the DP constraint in (2) is satisfied.

Minimum mixing weight (details in Theorem 4.1):

Given a noise distribution P_{nse} , privacy parameters ϵ, δ , number of synthetic samples m , and private dataset size n , there exists a threshold β_{min} such that any $\beta \geq \beta_{min}$ satisfies (ϵ, δ) -DP. Moreover,

- 1) β_{min} is a function of ϵ, δ, m, n and P_{nse} .
- 2) β_{min} is increasing in m .
- 3) β_{min} depends on P_{nse} only through its two smallest probability masses.

Intuition: For given parameters $\epsilon, \delta, m, n, P_{nse}$, the privacy constraint in (2) can be equivalently written as:

$$E_{e^\epsilon}(P_{syn}^{\otimes m} \| P_{syn}'^{\otimes m}) \leq \delta, \quad \forall D_{prv}, D'_{prv} \quad (4)$$

where $E_{e^\epsilon}(P \| Q)$ denotes the hockey-stick divergence between distributions P and Q (or the E_γ divergence with $\gamma = e^\epsilon$), and $P^{\otimes m}$ denotes the m -fold product distribution corresponding to m i.i.d. draws of P . Solving (4) with equality yields the minimum mixing weight β_{min} for the chosen P_{nse} . Importantly, since neighboring private datasets differ in exactly one sample, the ratio $\frac{P_{syn}}{P'_{syn}}$ is exactly 1 in all but (at most) two coordinates. Leveraging this 2-point sparse structure tightens the privacy accounting compared to existing DP histogram release methods.

For a given P_{nse} , the value of β_{min} specifies the minimum separation that P_{syn} must maintain from $\bar{P}_{prv}$ to satisfy DP. Since the information leakage increases with the number of synthetic samples released, β_{min} increases with m (see Fig. 2). The next property that β_{min} depends on P_{nse} only through its two smallest probability masses is also intuitive: Since the same mixing weight β is applied to every coordinate in (1), the coordinates with the smallest noise masses in P_{nse} receive the least added randomness. The worst case therefore occurs when the two coordinates that differ in neighboring private histograms coincide with the indices of the two smallest entries of P_{nse} . Consequently, letting $N_{min_1} = \min_i P_{nse}(i)$ and $N_{min_2} = \min_{i \neq i_1} P_{nse}(i)$ where $i_1 = \arg \min_i P_{nse}(i)$, the pair (N_{min_1}, N_{min_2}) determines the smallest β for which DP is satisfied. Since β_{min} depends on P_{nse} only through its two smallest probability masses N_{min_1} and N_{min_2} , we write $\beta_{min} = \beta_{min}(N_{min_1}, N_{min_2})$ to make it explicit.

Consider $N_{min_1} = N_{min_2} = N_{min}$. The behavior of $\beta_{min}(N_{min}, N_{min})$ for $0 \leq N_{min} \leq \frac{1}{d}$ is shown in Fig. 2, where $d = |\mathcal{X}|$. $\beta_{min}(N_{min}, N_{min})$ is decreasing in N_{min} as a larger value of N_{min} corresponds to a larger baseline noise being injected across all coordinates of $\bar{P}_{prv}$. Since more noise is already guaranteed at each coordinate, a smaller mixing weight β suffices to satisfy DP.

Having characterized the minimum mixing weight for a fixed noise distribution P_{nse} , we next optimize over P_{nse} .

Asymptotically optimum noise distribution (details in Theorem 4.2): For a given public distribution $\bar{P}_{pub}$ and $\gamma > 0$, the asymptotically optimum noise distribution P_{nse}^* , characterized in Theorem 4.2, is a *floor-raised* version of $\bar{P}_{pub}$ with an optimal threshold t . Floor-raising a distribution $\bar{P}_{pub}$ with threshold t raises all coordinates below t at least up to t , and compensates by reducing the mass from coordinates above t in a way that preserves normalization (see examples in Fig. 3). Moreover, the optimum threshold t is a function of $\gamma, \bar{P}_{pub}, \varepsilon, \delta, m$ and n .

Intuition: To determine the optimum noise distribution, we first observe that $\text{TV}(\bar{P}_{prv}, P_{syn}) = \beta \text{TV}(\bar{P}_{prv}, P_{nse})$. Then, the optimization in (3) can be upper bounded by:

$$\min_{P_{nse} \in \Delta_d} \beta_{min}(N_{min_1}, N_{min_1}) \left(\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) \right) \quad (5)$$

since the mixing weight β is chosen independently of $\bar{P}_{prv}$ and $\beta_{min}(N_{min_1}, N_{min_2})$ is decreasing in N_{min_2} . The optimal P_{nse}^* in (5) is an asymptotic optimizer for (3) since (5) converges to (3) as $d \rightarrow \infty$. Details in Sec. 4.

Consider the two terms in (5). $\beta_{min}(N_{min_1}, N_{min_1})$ is minimized when N_{min_1} is maximized (see Fig. 2). Thus, the P_{nse} minimizing the first term is *the uniform distribution*, where $N_{min_1} = \frac{1}{d}$. Now consider the second term (inner max). For any given P_{nse} , the inner max is achieved by some $\bar{P}_{prv}$ on the boundary of $\mathcal{B}_{\gamma, \bar{P}_{pub}}$. Therefore, the minmax TV distance is achieved by the P_{nse} at the L_1 -Chebyshev center³ of $\mathcal{B}_{\gamma, \bar{P}_{pub}}$ (close to $\bar{P}_{pub}$). The first term pushes P_{nse} toward the center of the simplex Δ_d (the uniform distribution), while the second term pushes it toward the Chebyshev center of $\mathcal{B}_{\gamma, \bar{P}_{pub}}$, which lies close to $\bar{P}_{pub}$. The optimal P_{nse}^* balances these two effects: it stays close to $\bar{P}_{pub}$ to minimize the worst-case TV term, while simultaneously increasing N_{min_1} to reduce the $\beta_{min}(N_{min_1}, N_{min_1})$ term. In Theorem 4.2, we show that the resulting optimum P_{nse}^* is a *floor-raised* version of $\bar{P}_{pub}$ characterized by a linear program. Moreover, we show that P_{nse}^* can be approximated by one of two easily computable floor-raising mechanisms explained in Fig. 3. These floor-raising operations slightly move $\bar{P}_{pub}$ towards the uniform distribution (details in App. B), thereby increasing N_{min_1} and lowering $\beta_{min}(N_{min_1}, N_{min_1})$. At the same time, it preserves proximity to the minimizer of the inner maximization term.

Significance of public data: Consider the asymptotic solu-

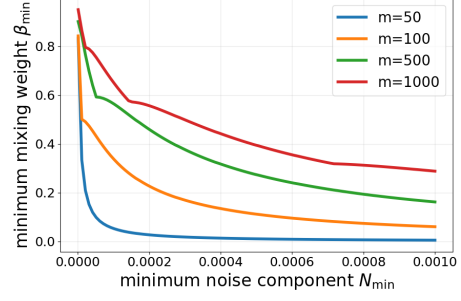


Figure 2. Comparison of minimum mixing weight β_{min} for different numbers of synthetic samples m and minimum noise components N_{min} with $n = 1200, d = 1000, \varepsilon = 5$ and $\delta = 1/n$.

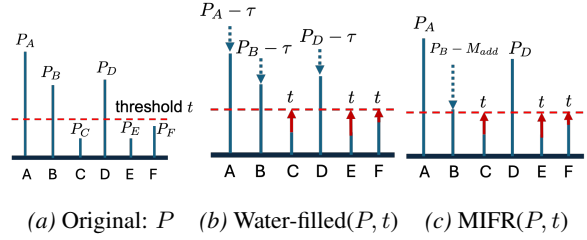


Figure 3. Two instances of floor-raising: (a) original PMF, (b) Water-filling: raises smaller masses to t and removes the added mass uniformly from larger masses (Def. 4.3), (c) Maximum Invariant Floor Raising (MIFR): raises smaller masses to t and removes the added mass M_{add} from larger ones to maximize the unchanged probability mass with respect to the original (Def. 4.4).

tion to (3) when no public data is available, i.e., $\mathcal{B}_{\gamma, \bar{P}_{pub}} = \Delta_d$. For this, we again analyze the two terms in (5) separately. The inner maximization in (5) is attained by a $\bar{P}_{prv}$ at an extreme point of the simplex (e.g., a corner of the triangle in Fig. 1). Thus, the noise distribution minimizing the inner maximum term in (5), i.e., the minmax optimal distribution, is the uniform distribution, which is at the center of the simplex. Now, consider the $\beta_{min}(N_{min_1}, N_{min_1})$ term in (5). Since $\beta_{min}(N_{min_1}, N_{min_1})$ is decreasing in N_{min_1} , it is minimized when $N_{min_1} = \frac{1}{d}$. Both of these facts imply that the objective in (5) is minimized by $P_{nse}^* = \frac{1}{d} \mathbf{1}_d$, i.e., the uniform distribution (see Corollary 4.6). We call this mechanism LINMIX.

However, this solution is driven by the worst-case private distribution, which is a point mass at one of the corners of the simplex. A point mass is a highly atypical and practically irrelevant distribution. Optimizing P_{nse}^* based on a point mass forces P_{nse}^* to be far from most realistic private distributions $\bar{P}_{prv}$, which results in a large TV distance between the private and synthetic distributions, as shown in Fig. 1. To mitigate this, we leverage public data from the same domain and solve (3) with $\mathcal{B}_{\gamma, \bar{P}_{pub}} \subset \Delta_d$. This results in PUBMIX, which achieves significantly improved performance (red curves).

³ L_1 -Chebyshev center of a set Q : $\arg \min_c \max_{x \in Q} \|x - c\|_1$.

Algorithm 1 PUBMIX

Input: : Private distribution $\bar{P}_{prv}$, public distribution $\bar{P}_{pub}$, private dataset size n , required sample size m , proximity parameter γ , privacy parameters (ε, δ)

Output: : Synthetic dataset D_{syn}

- 1: Compute near-optimal noise distribution P_{nse}^* and mixing weight β^* using $\bar{P}_{pub}, \gamma, \varepsilon, \delta, m, n$ (Remark 4.5)
- 2: Form the synthetic distribution:

$$P_{syn} \leftarrow (1 - \beta^*) \bar{P}_{prv} + \beta^* P_{nse}^*$$

- 3: Draw m i.i.d. samples from P_{syn} to obtain D_{syn}
- 4: **return** D_{syn}

3.2. Beyond Histograms

PUBMIX extends naturally to general discrete distribution estimators that satisfy a bounded sensitivity condition, namely, $\max_i |\bar{P}_{prv}^{[i]} - \bar{P}'_{prv}^{[i]}| \leq s$ for some $s > 0$, where $\bar{P}_{prv}^{[i]}$ denotes the i -th coordinate of $\bar{P}_{prv}$. In this setting, the domain-aware noise distribution continues to take the form of a floor-raised version of $\bar{P}_{pub}$, while the minimum mixing weight β is increased as we lose the two-point sparsity structure of neighboring histograms (see Appendix H). The same principles suggest potential extensions of PUBMIX to parametric distribution families, such as Gaussian mixtures.

4. Main Results: Technical Details

In this section, we formalize the arguments made in Sec. 3. We begin by deriving a condition on the mixing weight β that guarantees DP for a given noise distribution P_{nse} .

Theorem 4.1 (Feasible mixing weights β). *Fix $\varepsilon > 0$, $\delta \in (0, 1)$, private dataset size n , and the required number of synthetic samples $m \geq 1$. Let $P_{nse} \in \Delta_d$ be any noise distribution and denote the two smallest probability masses of P_{nse} by N_{min_1} and N_{min_2} , with $N_{min_1} \leq N_{min_2}$. Then, for every $\beta \geq \beta_{min}(N_{min_1}, N_{min_2})$, the m synthetic samples drawn from P_{syn} in (1) satisfy the DP constraint in (2), where $\beta_{min}(N_{min_1}, N_{min_2})$ is the solution to β in:*

$$\max\{h_1(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2}), h_2(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2})\} = \delta \quad (6)$$

where

$$\begin{aligned} h_1(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2}) &= \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} \left(1 - \beta(N_{min_1} + N_{min_2}) - \frac{1 - \beta}{n}\right)^{m - \lambda - \mu} \\ &\quad \left[\left(\beta N_{min_1} + \frac{1 - \beta}{n}\right)^\lambda (\beta N_{min_2})^\mu - e^\varepsilon \left(\beta N_{min_2} + \frac{1 - \beta}{n}\right)^\mu (\beta N_{min_1})^\lambda \right] + \end{aligned} \quad (7)$$

with $[x]_+ = \max\{x, 0\}$, and $h_2(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2})$ is the same as $h_1(\cdot)$ with N_{min_1} and N_{min_2} swapped.

The proof of Thm. 4.1 is given in App. C. Since the mixing weight β has to be as small as possible for optimal utility (Fig. 1a), for any given P_{nse} , the optimum mixing weight is the corresponding $\beta_{min}(N_{min_1}, N_{min_2})$ from Thm. 4.1.

Next, we analyze the optimal noise distribution P_{nse}^* . For this, we provide an asymptotically optimal solution to (3) in Thm. 4.2. The original problem in (3) is equivalently to:

$$\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_2})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn})$$

which can be bounded as:

$$\begin{aligned} &\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_2})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \\ &\leq \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \end{aligned} \quad (8)$$

as $N_{min_1} \leq N_{min_2}$ and $\beta(N_{min_1}, N_{min_2})$ is non-increasing in both N_{min_1} and N_{min_2} (see Lemma C.5). In Theorem 4.2, we provide the solution to the upper bound in (8). Notice that $\sum_i P_{nse}^{[i]} = 1$ requires:

$$0 \leq N_{min_1} \leq N_{min_2} \leq \frac{1 - N_{min_1}}{d - 1} \quad (9)$$

which makes $N_{min_2} \rightarrow N_{min_1}$ for larger d , resulting in $\beta_{min}(N_{min_1}, N_{min_2}) \rightarrow \beta_{min}(N_{min_1}, N_{min_1})$. Thus, the solution to the upper bound in (8) is asymptotically optimal for (3) (see App. D.1 for rigorous proof).

Theorem 4.2 (Asymptotically optimal parameters:). *Let $\gamma \in (0, 1]$ be a constant and let $\bar{P}_{pub} \in \Delta_d$ be a public distribution. Fix the privacy parameters $\varepsilon > 0$, $\delta \in (0, 1)$ and the required number of synthetic samples $m \geq 1$. Then,*

$$\begin{aligned} &\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \\ &= \min_{0 \leq t \leq \frac{1}{d}} \beta_{min}(t, t) f(t) \end{aligned} \quad (10)$$

where $f(t)$ is the solution to the following linear program:

$$\begin{aligned} &f(t) = \min_{P_{nse} \in \Delta_d, r \in \mathbb{R}} r \\ &\text{s.t. } r + P_{nse}(S) \geq \min\{\bar{P}_{pub}(S) + \gamma, 1\}, \\ &\quad \forall S \subset [d], \quad 1 \leq |S| \leq d - 1 \\ &\quad P_{nse}^{[i]} \geq t, \forall i, \quad \mathbf{1}_d^T P_{nse} = 1 \end{aligned} \quad (11)$$

with the notation $P(S) = \sum_{i \in S} P^{[i]}$. The corresponding optimum P_{nse}^* is the minimizing P_{nse} of (11) with $t = t^*$ where $t^* = \arg \min_{0 \leq t \leq \frac{1}{d}} \beta_{min}(t, t) f(t)$. The optimum β^* is given by $\beta^* = \beta_{min}(t^*, t^*)$.

The proof of Theorem 4.2 is given in App. E. We next develop a (sub-optimal) approximation for P_{nse}^* in (11), on which PUBMIX is built. For this, we first define *Water-filling* and *Maximum-invariance floor raising*.

Definition 4.3. [Water-Filling (WF)] Let A be a probability mass function (PMF) on $\Omega = \{\omega_1, \dots, \omega_d\}$ with $A_i := A(\omega_i)$. Assume without loss of generality that the indices are arranged such that $A_1 \leq A_2 \leq \dots \leq A_d$. Let $t \in [0, 1/d]$ be any constant satisfying $A_1 \leq \dots \leq A_k \leq t < A_{k+1} \leq \dots \leq A_d$ for some $k < d$. Then, the t -water-filled version of A , denoted by $B = \text{WF}(A, t)$ is given by,

$$B_i = \max\{t, A_i - \tau\}, \quad \forall i \in \{1, \dots, d\} \quad (12)$$

where $\tau = \frac{t(k+\ell) - \sum_{i=1}^{k+\ell} A_i}{d-k-\ell}$ is a unique constant satisfying $\sum_{i=1}^d B_i = 1$ and $A_{k+\ell} - t \leq \tau < A_{k+\ell+1} - t$ for a unique $\ell \in \{0, 1, \dots, d-k-1\}$.

Definition 4.4. [Maximum-Invariance Floor Raising (MIFR)] Let A be a probability mass function on $\Omega = \{\omega_1, \dots, \omega_d\}$ with $A_i := A(\omega_i)$. Fix $t \in [0, 1/d]$. The *maximum-invariance floor raised* version of A at level t is given by $B = \text{MIFR}(A, t)$ with coordinates

$$B_i = \begin{cases} t, & i \in L, \\ b_i, & i \in U^*, \\ A_i, & i \notin L \cup U^*, \end{cases}$$

where $L := \{i : A_i \leq t\}$ and $L^c := \{i : A_i > t\}$. The donor set $U^* \subseteq L^c$ is defined as

$$U^* = \arg \min_{U \subseteq L^c} \sum_{i \in U} A_i \quad \text{s.t.} \quad \sum_{i \in U} (A_i - t) \geq \sum_{i \in L} (t - A_i).$$

Finally, the values $\{b_i\}_{i \in U^*}$ satisfy $b_i \geq t, \forall i \in U^*$ and

$$\sum_{i \in U^*} (A_i - b_i) = \sum_{i \in L} (t - A_i),$$

so that the total mass removed from indices in U^* exactly matches the total mass added on L .

Both WF and MIFR (Def. 4.3 and 4.4) raise every probability mass below the threshold t up to t . The extra mass added in this step is then removed from entries above t , while ensuring no entry is reduced below t . In this way, both procedures produce a modified PMF B that satisfies $B_i \geq t$ and attains the minimum possible TV distance to the original PMF A . The two methods differ in how they choose which entries above t to decrease. WF spreads the required mass removal across all entries above t in a uniform “leveling” manner until some reach t . In contrast, MIFR concentrates the mass removal on a carefully chosen subset of entries above t so as to keep as many coordinates unchanged as possible, i.e., it maximizes the invariant mass $\sum_{i: B_i = A_i} A_i$. An illustration of Def. 4.3 and Def. 4.4 is given in Fig. 3.

Remark 4.5. For given parameters $\varepsilon, \delta, m, n, \bar{P}_{pub}, \gamma$, an approximation to the P_{nse}^* in Theorem 4.2 is given by,

$$\hat{P}_{nse}^* = \begin{cases} \text{MIFR}(\bar{P}_{pub}, t_1), & g(\text{MIFR}(\bar{P}_{pub}, t_1)) \leq g(\text{WF}(\bar{P}_{pub}, t_2)) \\ \text{WF}(\bar{P}_{pub}, t_2), & \text{o.w.} \end{cases}$$

where t_1, t_2 are optimized thresholds, and

$$g(P_{nse}) = \beta_{\min}(N_{\min_1}, N_{\min_1}) \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse})$$

for any P_{nse} with $N_{\min_1} = \min_i P_{nse}^{[i]}$. The corresponding minimum mixing weight is given by,

$$\hat{\beta}^* = \beta_{\min}(N_{\min_1}^*, N_{\min_1}^*)$$

where $N_{\min_1}^* = \min_i \hat{P}_{nse}^{*[i]}$. In most cases, $\hat{P}_{nse}^*$ coincides with P_{nse}^* from Theorem 4.2, but it is generally sub-optimal (see App. F for more details).

The **Key insight from Remark 4.5** is that $\hat{P}_{nse}^*$ is always a floor-raised version of $\bar{P}_{pub}$, where both $\text{MIFR}(\bar{P}_{pub}, t_1)$ and $\text{WF}(\bar{P}_{pub}, t_2)$ raise small elements of $\bar{P}_{pub}$ to their respective thresholds and redistribute probability mass from larger elements. This design balances two objectives: since $\bar{P}_{prv}$ and $\bar{P}_{pub}$ come from the same domain with typically small $\text{TV}(\bar{P}_{pub}, \bar{P}_{prv})$, choosing P_{nse} close to $\bar{P}_{pub}$ is beneficial. However, many elements in $\bar{P}_{pub}$ can be small (even zero), which would add minimal noise to corresponding $\bar{P}_{prv}^{[i]}$ elements and require a larger mixing weight β to maintain privacy, pushing P_{syn} too close to $\bar{P}_{pub}$ (not to $\bar{P}_{prv}$). $\hat{P}_{nse}^*$ resolves this tradeoff by remaining close to $\bar{P}_{pub}$ while floor-raising smaller probabilities to guarantee a sufficient $N_{\min_1} = \min_i \hat{P}_{nse}^{[i]}$, which allows a smaller β and yields a P_{syn} that is closer to $\bar{P}_{prv}$.

To see the significance of using domain information via public data, consider the solution to (3) when no public dataset D_{pub} is available, i.e., when $\mathcal{B}_{\gamma, \bar{P}_{pub}} = \Delta_d$.

Corollary 4.6 (Asymptotic optimality with no D_{pub}). *For any D_{prv} , $m \geq 1$, $\varepsilon > 0$ and $\delta \in (0, 1)$, the asymptotic solution to (3) with $\mathcal{B}_{\gamma, \bar{P}_{pub}} = \Delta_d$ is given by,*

$$\begin{aligned} \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \Delta_d} \text{TV}(\bar{P}_{prv}, P_{syn}) \\ = (1 - 1/d) \beta_{\min}(1/d, 1/d) \end{aligned} \quad (13)$$

and the corresponding optimum P_{nse}^* and β^* are given by,

$$P_{nse}^* = \frac{1}{d} \mathbf{1}_d \quad \text{and} \quad \beta^* = \beta_{\min}(1/d, 1/d) \quad (14)$$

where $\mathbf{1}_d$ is the all ones vector of size d .

Corollary 4.6 (proof in App. G) shows that without domain information (i.e., no public data from the same domain as D_{prv}), the optimal noise distribution is uniform and P_{syn} is formed by mixing uniform noise with $\bar{P}_{prv}$. We call this mechanism LINMIX. This effectively pulls P_{syn} toward the uniform distribution and away from $\bar{P}_{prv}$. In contrast, when relevant public data is available, Theorem 4.2 and Remark 4.5 establish that the optimal noise distribution is close to $\bar{P}_{pub}$. Since $\bar{P}_{pub}$ and $\bar{P}_{prv}$ share similar structural properties, linearly combining them preserves much of this structure, allowing P_{syn} to remain closer to $\bar{P}_{prv}$ while still satisfying the privacy constraint.

5. Experiments

In the first part of this section, we illustrate the core ideas of PUBMIX with a simplified setting. In the second part, we integrate PUBMIX into state-of-the-art DP synthetic data generation pipelines (histogram-based) and show improved performance.⁴ First, we explain how the proximity parameter γ is chosen in both experimental settings. γ captures the variation of the domain data: if distributions estimated from different public datasets in the same domain vary substantially, γ should be larger. If they are similar, a smaller γ is appropriate. In our experiments, we either collect k public datasets from the same domain, or randomly partition the available public dataset into k folds, compute pairwise TV distances between the fold distributions, and use the distribution of these distances to estimate γ .

5.1. Illustrating the Core Concepts

In this section, we use a simplified setting in which the normalized histogram of the private data serves as the distribution estimate, and PUBMIX is used to directly sample synthetic data. To make this setting tractable, we use the 2023 ACS Public Use Microdata Sample (PUMS) dataset (U.S. Census Bureau, 2024) by restricting each record with 7 features: race, sex, class of work, marital status, education level, disability status, and English proficiency. We analyze the quality of the PUBMIX-generated samples in terms of TV distance and downstream-task performance by comparing with the standard Gaussian mechanism and the setting with no public data (LINMIX). We use Texas data as D_{prv} and California data as D_{pub} .⁵ Fig. 4 compares the TV distance between private distribution and the empirical distribution of the generated synthetic data. LINMIX performs well for small m but degrades as m increases, while PUBMIX avoids this degradation by leveraging public data.

We assess downstream utility by training a random forest re-

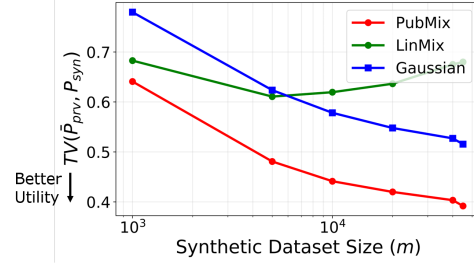


Figure 4. Comparison of the total variation (TV) distance between $\bar{P}_{prv}$ and the empirical distribution of the synthetic dataset D_{syn} for $n = 57,717$, $\epsilon = 5$, and $\delta = 1/n$ on the PUMS dataset.

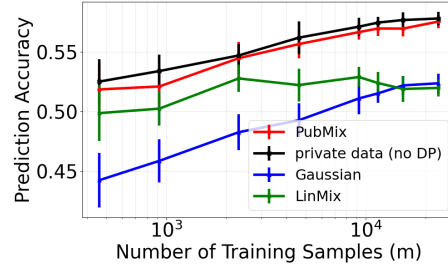


Figure 5. Accuracy of a regression model trained on synthetic data by PUBMIX, LINMIX, and the Gaussian mechanism, measured relative to a model trained on real data from the PUMS dataset.

gressor to predict *English proficiency* (values 1-5) using the remaining features, with the model trained on the generated synthetic samples. Here, we use 80% of Texas data as D_{prv} to generate the synthetic samples. We then use the remaining 20% of Texas data as the test set to evaluate the performance of the model trained on the synthetic data. Fig. 5 compares PUBMIX against a non-private model trained directly on D_{prv} , the Gaussian baseline, and LINMIX (no D_{pub}).

5.2. Evaluation on Existing Pipelines

We next evaluate PUBMIX on two modalities, tabular and text, to demonstrate that it can be used as a *drop-in* replacement for the DP histogram sampling subroutine in existing state-of-the-art synthetic data generation pipelines. Concretely, we adopt Private Evolution (PE) based methods (Tran et al., 2026; Xie et al., 2024), which are recently proposed frameworks generating synthetic data through an iterative process: they repeatedly produce candidate variations of the data and then randomly samples from Gaussian-noised histograms obtained from the variations. In our experiments, we keep the PE pipeline unchanged and replace only the internal DP histogram-sampling component with PUBMIX.

5.2.1. Tabular: Person Activity Records

Experimental setup. We use the Person Activity dataset (Vidulin et al., 2010), which contains sensor-based activity records collected from five users. To mimic a realistic deployment where related public data is available, we con-

⁴Code available at: <https://github.com/sajani-vithana/PUBMIX>

⁵We set γ to the average pairwise TV distance among state-level distributions from California, Colorado, Florida, and Utah.

Table 1. Downstream classification accuracy (%) on Person Activity under varying privacy budgets. We report mean $\pm$ std.

Method	$\epsilon = 1.0$	$\epsilon = 2.0$	$\epsilon = 4.0$
GEM (Liu et al., 2021b)	28.24 $\pm$ 0.43	28.51 $\pm$ 1.84	28.46 $\pm$ 0.40
PrivSyn (Zhang et al., 2021)	27.40 $\pm$ 2.70	28.41 $\pm$ 2.03	28.06 $\pm$ 1.82
GSD (Liu et al., 2023)	49.48 $\pm$ 0.27	50.00 $\pm$ 0.25	50.46 $\pm$ 0.54
AIM (McKenna et al., 2022)	56.77 $\pm$ 0.23	58.42 $\pm$ 0.11	59.12 $\pm$ 0.30
JAM-PGM (Fuentes et al., 2024)	55.96 $\pm$ 0.24	56.51 $\pm$ 0.28	57.37 $\pm$ 0.23
PMW _{pub} (Liu et al., 2021a)	49.21 $\pm$ 0.80	48.96 $\pm$ 1.05	49.37 $\pm$ 0.91
TABPE (Tran et al., 2026)	60.80 $\pm$ 0.54	63.22 $\pm$ 0.48	64.24 $\pm$ 0.45
PUBMIX	66.40$\pm$0.49	66.47$\pm$0.42	66.85$\pm$0.08

struct the private dataset D_{prv} using records from three users, and the public dataset D_{pub} using records from the remaining two users by splitting the original dataset. This split preserves the domain and task semantics while introducing a natural distribution shift across users, matching the intended use of public-data-guided sample generation.

As our primary baseline, we adopt TabPE (Tran et al., 2026), a state-of-the-art DP synthetic data generator for tabular data. Importantly, TABPE relies on a DP histogram-based sampling subroutine as a core primitive in its update procedure. Building on this modular structure, our main comparison is TABPE+PUBMIX, where we *only* replace this DP histogram sampling component in TABPE with PUBMIX while keeping all other components (e.g., model architecture, iterations, and optimization settings) unchanged. This experimental design isolates the benefit of PUBMIX as a drop-in sampling primitive. We also compare PUBMIX against additional tabular DP baselines, including GSD (Liu et al., 2023), AIM (McKenna et al., 2022), GEM (Liu et al., 2021b), PRIVSYN (Zhang et al., 2021), as well as JAM-PGM (Fuentes et al., 2024) and PWM_{pub} (Liu et al., 2021a) that utilize public data. Following prior work (Tran et al., 2026), we evaluate synthetic data utility by training a downstream classifier on synthetic samples and reporting test accuracy of classification performance for human activity.

Results. Table 1 reports downstream accuracy across privacy budgets. Across all $\epsilon \in \{1, 2, 4\}$, TABPE + PUBMIX achieves the best performance, consistently outperforming the PE baseline as well as all other DP tabular methods. Notably, the gains over PE are largest in the stricter privacy regime: PE+PUBMIX improves PE by +5.80 points at $\epsilon = 1$ (66.60 vs. 60.80), and remains advantageous at $\epsilon = 2$ and $\epsilon = 4$ (+2.93 and +1.88 points, respectively). This pattern shows that PUBMIX leverages related public data to improve utility under stronger privacy constraints.

5.2.2. Text: Yelp Reviews

Experimental setup. We consider a real-world text synthesis setting where the private dataset is Yelp reviews (Yelp Inc., 2023), and the public dataset is Google Local Reviews from 10 states (Li et al., 2022; Yan et al., 2023). This public dataset provides a realistic public signal from a related

Table 2. Utility on Yelp Reviews under varying privacy budgets. We report mean $\pm$ std over multiple runs. Category classification accuracy (%), higher is better) and Rating RMSE (lower is better). PE ($\epsilon = \infty$): Category 72.13%, RMSE 0.873.

Metric	Method	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$
Category ($\uparrow$)	PE (Xie et al., 2024)	71.67$\pm$0.21	70.69 $\pm$ 0.41	69.93 $\pm$ 0.38
	PUBMIX	71.01 $\pm$ 0.17	71.57$\pm$0.34	72.35$\pm$0.20
Rating ($\downarrow$)	PE (Xie et al., 2024)	0.90 $\pm$ 0.025	0.91 $\pm$ 0.009	0.89 $\pm$ 0.021
	PUBMIX	0.81$\pm$0.016	0.81$\pm$0.022	0.80$\pm$0.039

but non-identical distribution, reflecting practical scenarios where public web-scale reviews exist but do not exactly match the private domain. Yelp provides two downstream tasks that capture complementary aspects of utility. (i) Category classification: a trained classifier on the synthetic reviews predicts the business category. (ii) Rating prediction: a trained regressor predicts the star rating in $\{1, \dots, 5\}$. We report classification accuracy and the root mean squared error (RMSE) for the rating, on the test set.

We adopt a PE variant method, tailored for DP text synthesis as our main baseline (Xie et al., 2024). We apply PUBMIX as a plug-in replacement in the DP sampling component of the PE pipeline, while keeping all remaining settings identical. Additional details (length constraints, model configuration, and training schedule) are provided in App. I.

Results. Table 2 summarizes utility on Yelp across privacy budgets. Overall, PUBMIX improves the privacy-utility tradeoff over the PE baseline on both tasks. In particular, PUBMIX consistently reduces rating prediction error (RMSE) for all ϵ , indicating better preservation of the continuous rating signal under DP. These results suggest that public-guided DP sampling can enhance PE-based text synthesis without changing the rest of the pipeline.

6. Conclusion

In this paper, we introduced the concept of domain-aware DP mechanisms for synthetic data generation and provided a corresponding theoretical framework. For normalized histograms, we characterized the optimal domain-aware DP mechanism within a class of distribution mixing functions. To this end, we introduced PUBMIX, a public-data-aware privacy mechanism that can be used as a drop-in replacement for Laplace or Gaussian mechanisms in histogram-based data synthesis pipelines. Empirically, we show that incorporating PUBMIX in existing data synthesis pipelines consistently improves the privacy-utility trade-off.

Limitations and future work: Our analysis is restricted to linear mixing-based DP mechanisms, requires public data from the same domain, and considers only histogram-based distribution estimators. Future work will extend domain-aware privacy mechanisms to more complex distribution estimators, broadening the scope of PUBMIX’s applications.

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Impact Statement

This work advances DP synthetic generation by proposing public-data-aware privacy mechanisms. This method incorporates auxiliary public data directly into the privacy-critical sampling step, improving utility while maintaining DP guarantees. By enabling the generation of higher-fidelity synthetic data under a given privacy budget, the proposed approach increases the practicality of privacy-preserving data sharing in domains where data access is limited (e.g., healthcare, finance and user-generated content). At the same time, synthetic data can be misused or misinterpreted. First, since DP protections depend on correct choice and reporting of privacy parameters, loose privacy budgets can undermine protections. Second, synthetic data may still encode and amplify social biases present in the private dataset. Finally, because our method leverages public data, careful governance is needed to ensure the public source is legitimately shareable and properly licensed.

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A. Additional Details on Related Work

On methods using linear mixing: Certain private prediction variants (Flemings et al., 2024; Ginart et al., 2022) are alternatives to DP training methods such as DP stochastic gradient descent (DP-SGD), where instead of privatizing the model itself, privacy is enforced at inference time by perturbing the model’s output. In the LLM setting, this typically involves mixing or projecting the private model’s next-token distribution toward a public distribution to limit leakage from memorized private data. While both PubMix and private prediction variants involve linear mixing and public distributions, they differ fundamentally in goal and technical design.

- **Goal:** Private prediction protects the privacy of an LLM’s fine-tuning data at inference time, one token prediction at a time. In contrast, PUBMIX introduces domain-aware, sample-aware synthetic data generation from histograms, with the goal of producing m privacy-protected samples whose distribution is as close as possible to the private data distribution.
- **Optimization:** Private prediction methods (Flemings et al., 2024; Ginart et al., 2022) linearly mix private token distributions with a fixed public distribution and optimize only the mixing coefficient to remain as close as possible to the private distribution while satisfying privacy. PUBMIX instead mixes a private normalized histogram with a noise distribution and jointly optimizes both the mixing weight β and the noise distribution P_{nse} to minimize the distance to the private distribution. In this sense, PubMix generalizes the optimization step of private prediction by introducing an additional optimization variable, which provides extra flexibility to improve the utility of the mixing step.
- **Role of public data:** In private prediction, the public distribution mainly serves as a source of noise. In PUBMIX, public data is used more structurally: it informs the design of a domain-aware noise distribution. In other words, using the public distribution as the noise distribution is only a special case of PUBMIX. Generally, the optimal noise distribution drifts toward a floor-raised version of the public distribution. This distinction matters especially when the public distribution is sparse (which is common in token distributions), since fixing it directly can force the mixed distribution to be much closer to the public distribution than to the private ones, which degrades utility. PUBMIX avoids this by jointly optimizing the noise distribution and mixing parameter.

While these differences exist between PUBMIX and private prediction methods, the extension of PUBMIX to general discrete distributions beyond histograms (briefly discussed in App. H) can be applied on token distributions. In that sense, PUBMIX can potentially serve as a drop-in privacy module for the linear mixing step in private prediction, provided it is paired with an appropriate privacy accounting across generated tokens.

B. Proofs of Section 3

Lemma B.1. *Let $A = (A_1, \dots, A_d)$ be a probability mass function (PMF) on a finite set $\Omega = \{\omega_1, \dots, \omega_d\}$, and let $U = (U_1, \dots, U_d)$ denote the uniform distribution on Ω , i.e., $U_i = 1/d$ for all i . For any threshold $t \in [0, 1/d]$, let $B = \text{WF}(A, t)$ denote the t -water-filled version of A as in Definition 4.3. Then, the total variation distance to the uniform distribution does not increase under water-filling, i.e., $\|B - U\|_{\text{TV}} \leq \|A - U\|_{\text{TV}}$.*

Proof. Let $A = (A_1, \dots, A_d)$ be the original PMF, $B = (B_1, \dots, B_d)$ the water-filled PMF, and $U = (U_1, \dots, U_d)$ the uniform distribution with $U_i = 1/d$. We also denote by $Q = (Q_1, \dots, Q_d)$ an arbitrary PMF for intermediate arguments. For any PMF Q on Ω , the total variation distance to the uniform distribution U satisfies

$$\|Q - U\|_{\text{TV}} = \frac{1}{2} \sum_{i=1}^d |Q_i - \frac{1}{d}| = \sum_{i: Q_i > \frac{1}{d}} \left(Q_i - \frac{1}{d} \right). \quad (15)$$

Therefore, it suffices to show that

$$\sum_{i: B_i > \frac{1}{d}} \left(B_i - \frac{1}{d} \right) \leq \sum_{i: A_i > \frac{1}{d}} \left(A_i - \frac{1}{d} \right). \quad (16)$$

Recall that $B_i = \max\{t, A_i - \tau\}$, where τ is chosen such that $\sum_{i=1}^d B_i = 1$. We first show that $\tau \geq 0$. Suppose, for the sake of contradiction, that $\tau < 0$. Then $A_i - \tau > A_i$ for all i , which implies $B_i \geq A_i - \tau > A_i$ for all i . Consequently, $\sum_{i=1}^d B_i > \sum_{i=1}^d A_i = 1$, which contradicts the normalization condition $\sum_{i=1}^d B_i = 1$. Hence, $\tau \geq 0$.

Consider any index i such that $A_i \leq \frac{1}{d}$. Since $t \leq \frac{1}{d}$ and $\tau \geq 0$, we have

$$B_i = \max\{t, A_i - \tau\} \leq \frac{1}{d}. \quad (17)$$

Hence, no coordinate whose original mass is at most $1/d$ can exceed $1/d$ after water-filling.

Consider any index i such that $A_i > \frac{1}{d}$. As $A_i > t$ and $\tau \geq 0$,

$$B_i = A_i - \tau \leq A_i. \quad (18)$$

Hence, any excess mass above $1/d$ is weakly reduced by the water-filling operation.

Combining the above observations, the set $\{i : B_i > \frac{1}{d}\}$ is a subset of $\{i : A_i > \frac{1}{d}\}$, and for each such index, $B_i - \frac{1}{d} \leq A_i - \frac{1}{d}$. Summing over all indices yields

$$\|B - U\|_{\text{TV}} \leq \|A - U\|_{\text{TV}}, \quad (19)$$

which completes the proof. $\square$

C. Proof of Theorem 4.1

In this section, we provide the proof of Theorem 4.1. First, we restate Theorem 4.1, and then provide the proof in the following order.

- Rewrite the privacy constraint in (2) in terms of the privacy parameters ε, δ , required samples m , noise distribution P_{nse} , mixing coefficient β , and $a = P'_{syn}(i)$, $b = P'_{syn}(j)$, where i and j are the two distinct indices at which the normalized histograms, $\bar{P}_{prv}$ and $\bar{P}'_{prv}$, of neighboring datasets D_{prv} and D'_{prv} differ. Specifically, we show that the privacy constraint simplifies to:

$$\sup_{a,b} E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m}) \leq \delta \quad (20)$$

where P and Q are distributions given by $P = \left[a + \frac{1-\beta}{n}, b - \frac{1-\beta}{n}, 1-a-b\right]$ and $Q = [a, b, 1-a-b]$, and $E_{e^\varepsilon}(\cdot \| \cdot)$ is the E_γ divergence.

- We show that $\sup_{a,b} E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ is non-increasing in a and b for any given $\varepsilon, \delta, m, n, \beta, P_{nse}$, and characterize the exact $\sup_{a,b} E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$, which leads to Theorem 4.1.

Theorem 4.1 restated:[Feasible mixing weights β] Fix $\varepsilon > 0$, $\delta \in (0, 1)$, private dataset size n , and the required number of synthetic samples $m \geq 1$. Let $P_{nse} \in \Delta_d$ be any noise distribution and define N_{min_1} and N_{min_2} to be the two smallest elements of P_{nse} with $N_{min_1} \leq N_{min_2}$. Then, for every $\beta \geq \beta_{min}(N_{min_1}, N_{min_2})$, the m synthetic samples drawn from P_{syn} in (1) satisfy the DP constraint in (2), where $\beta_{min}(N_{min_1}, N_{min_2})$ is given by the solution to β in:

$$\max\{h_1(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2}), h_2(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2})\} = \delta \quad (21)$$

where

$$h_1(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2}) = \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} \left(1 - \beta(N_{min_1} + N_{min_2}) - \frac{1 - \beta}{n}\right)^{m - \lambda - \mu} \left[\left(\beta N_{min_1} + \frac{1 - \beta}{n}\right)^\lambda (\beta N_{min_2})^\mu - e^\varepsilon \left(\beta N_{min_2} + \frac{1 - \beta}{n}\right)^\mu (\beta N_{min_1})^\lambda \right]_+ \quad (22)$$

where $[x]_+ = \max\{x, 0\}$, and $h_2(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2})$ is the same as $h_1(\beta, \varepsilon, n, m, N_{min_1}, N_{min_2})$ with N_{min_1} and N_{min_2} swapped. Moreover, $\beta_{min}(N_{min_1}, N_{min_2})$ is decreasing in N_{min_1} and N_{min_2} .

Lemma C.1. For any fixed privacy parameters $\varepsilon > 0$ and $\delta \in (0, 1)$, required number of synthetic samples m , and any fixed noise distribution $P_{nse} \in \Delta_d$, the privacy constraint in (2) is simplified to:

$$\sup_{(a,b) \in \mathcal{F}(\beta, P_{nse})} E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m}) = \sup_{(a,b) \in \mathcal{F}(\beta, P_{nse})} f(a, b, \varepsilon, \beta) \leq \delta \quad (23)$$

where $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ denotes the hockey-stick divergence (E_γ divergence) between the product distributions $P^{\otimes m}$ and $Q^{\otimes m}$ with P and Q denoting the three-point distributions $P = [a + \frac{1-\beta}{n}, b - \frac{1-\beta}{n}, 1-a-b]$ and $Q = [a, b, 1-a-b]$. $\mathcal{F}(\beta, P_{nse})$ is the feasible set of (a, b) , and,

$$f(a, b, \varepsilon, \beta) = \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} (1 - a - b)^{m - \lambda - \mu} \left[\left(a + \frac{1 - \beta}{n} \right)^\lambda \left(b - \frac{1 - \beta}{n} \right)^\mu - e^\varepsilon a^\lambda b^\mu \right]_+ \leq \delta \quad (24)$$

Proof: Consider the privacy constraint in (2): $\mathbb{P}(D_{syn} \in S) \leq e^\varepsilon \mathbb{P}(D'_{syn} \in S) + \delta, \forall S \subseteq \mathcal{X}^m, \forall D_{prv}, D'_{prv}$, which is equivalent to:

$$\max_{S \subseteq \mathcal{X}^m} \mathbb{P}(D_{syn} \in S) - e^\varepsilon \mathbb{P}(D'_{syn} \in S) \leq \delta, \quad \forall D_{prv}, D'_{prv} \quad (25)$$

Let $\mathbf{Y} = \{y_1, \dots, y_m\} \in \mathcal{X}^m$ denote any generic candidate set of synthetic samples. Then, we can write the privacy constraint in (25) as:

$$\sum_{\mathbf{Y} \in \mathcal{X}^m} \left[\prod_{k=1}^m P_{syn}(y_k) - e^\varepsilon \prod_{k=1}^m P'_{syn}(y_k) \right]_+ \leq \delta, \quad \forall D_{prv}, D'_{prv} \quad (26)$$

where $[x]_+ = \max\{x, 0\}$. From the construction of P_{syn} in (1), we have,

$$P_{syn}(y) = (1 - \beta) \bar{P}_{prv}(y) + \beta P_{nse}(y) \quad (27)$$

$$P'_{syn}(y) = (1 - \beta) \bar{P}'_{prv}(y) + \beta P_{nse}(y) \quad (28)$$

for any $y \in \mathcal{X}$ where,

$$\bar{P}_{prv}(y) = \begin{cases} \bar{P}'_{prv}(y) + \frac{1}{n}, & y = i \\ \bar{P}'_{prv}(y) - \frac{1}{n}, & y = j \\ \bar{P}'_{prv}(y), & y \neq i, j \end{cases} \quad (29)$$

for some $i, j \in \mathcal{X}$. Since each y_k is sampled i.i.d. from P_{syn} (or P'_{syn}), (26) can be written in terms of the E_γ divergence between the product distributions $P_{syn}^{\otimes m}$ and $P'_{syn}^{\otimes m}$.

$$E_{e^\varepsilon}(P_{syn}^{\otimes m} \| P'_{syn}^{\otimes m}) \leq \delta, \quad \forall D_{prv}, D'_{prv} \quad (30)$$

For product distributions, the likelihood ratio of any realization $\mathbf{Y} = (y_1, \dots, y_m)$ is $\prod_{k=1}^m \frac{P_{syn}(y_k)}{P'_{syn}(y_k)}$, which depends only on the number of times i and j appear among the m samples, since $\frac{P_{syn}(y)}{P'_{syn}(y)} = 1$ for all $y \notin \{i, j\}$. Therefore, $E_{e^\varepsilon}(P_{syn}^{\otimes m} \| P'_{syn}^{\otimes m})$ depends on P_{syn} and P'_{syn} only through the probabilities assigned to i, j , and the total remaining mass. With the notation $P'_{syn}(i) = a, P'_{syn}(j) = b, P_{syn}(i) = a + \frac{1-\beta}{n}$ and $P_{syn}(j) = b - \frac{1-\beta}{n}$, we can equivalently write (30) as:

$$E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m}) \leq \delta, \quad \forall D_{prv}, D'_{prv} \quad (31)$$

where P and Q are three-point distributions given by $P = [a + \frac{1-\beta}{n}, b - \frac{1-\beta}{n}, 1 - a - b]$ and $Q = [a, b, 1 - a - b]$, corresponding to the probabilities of outcomes i, j , and the aggregate remainder in P_{syn} and P'_{syn} , respectively. Moreover, we can replace the $\forall D_{prv}, D'_{prv}$ term in (31) by the supremum over (a, b) for any given β and P_{nse} . This gives the following equivalent form of the privacy constraint in (31):

$$\sup_{(a,b) \in \mathcal{F}(\beta, P_{nse})} E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m}) \leq \delta \quad (32)$$

The numbers of samples “ i ”, “ j ”, or “anything other than i, j ” (sampled i.i.d.) follow a multinomial distribution with probabilities $(a + \frac{1-\beta}{n}, b - \frac{1-\beta}{n}, 1 - a - b)$ and $(a, b, 1 - a - b)$, respectively, for $P^{\otimes m}$ and $Q^{\otimes m}$. This simplifies the privacy constraint in (32) to:

$$\sup_{(a,b) \in \mathcal{F}(\beta, P_{nse})} \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} (1 - a - b)^{m - \lambda - \mu} \left[\left(a + \frac{1-\beta}{n} \right)^\lambda \left(b - \frac{1-\beta}{n} \right)^\mu - e^\varepsilon a^\lambda b^\mu \right]_+ \leq \delta \quad (33)$$

This completes the proof of Lemma C.1. ■

Lemma C.2. For any fixed $P_{nse} \in \Delta_d$ and $\beta \in [0, 1]$, $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ is non-increasing in a and b .

Proof: First, we show that $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ is non-increasing in a for any fixed b . For this, define $P_a = [a + \frac{1-\beta}{n}, b - \frac{1-\beta}{n}, 1 - a - b]$ and $Q_a = [a, b, 1 - a - b]$, and similarly $P_{a+\Delta} = [a + \Delta + \frac{1-\beta}{n}, b - \frac{1-\beta}{n}, 1 - a - \Delta - b]$ and $Q_{a+\Delta} = [a + \Delta, b, 1 - a - \Delta - b]$ for some small $\Delta > 0$. Consider the following stochastic channel T , applied independently to each sample y , sampled from P_a and Q_a :

$$T(y' | y) = \begin{cases} 1, & y' = y, \quad y \in \{i, j\}, \\ \frac{\Delta}{1 - a - b}, & y' = i, \quad y = \text{“other”}, \\ 1 - \frac{\Delta}{1 - a - b}, & y' = \text{“other”}, \quad y = \text{“other”}. \end{cases} \quad (34)$$

where “other” denotes the combined event of all $y \neq i, j$. The output distributions TP_a and TQ_a are given by:

$$(TP_a)(i) = a + \frac{1-\beta}{n} + \Delta, \quad (35)$$

$$(TP_a)(j) = b - \frac{1-\beta}{n}, \quad (36)$$

$$(TP_a)(\text{“other”}) = 1 - a - b - \Delta, \quad (37)$$

and

$$(TQ_a)(i) = a + \Delta, \quad (38)$$

$$(TQ_a)(j) = b, \quad (39)$$

$$(TQ_a)(\text{“other”}) = 1 - a - b - \Delta. \quad (40)$$

These match $P_{a+\Delta}$ and $Q_{a+\Delta}$, respectively:

$$TP_a = P_{a+\Delta} \quad \text{and} \quad TQ_a = Q_{a+\Delta}. \quad (41)$$

Since the channel T is applied to all m samples independently and identically, we have

$$T^{\otimes m} : P_a^{\otimes m} \mapsto P_{a+\Delta}^{\otimes m} \quad \text{and} \quad T^{\otimes m} : Q_a^{\otimes m} \mapsto Q_{a+\Delta}^{\otimes m}. \quad (42)$$

By the data processing inequality for the hockey-stick divergence (Sason & Verdú, 2016), which states that $E_\gamma(TP \| TQ) \leq E_\gamma(P \| Q)$ for any stochastic channel T , we conclude that

$$E_{e^\varepsilon}(P_{a+\Delta}^{\otimes m} \| Q_{a+\Delta}^{\otimes m}) \leq E_{e^\varepsilon}(P_a^{\otimes m} \| Q_a^{\otimes m}). \quad (43)$$

This establishes that $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ is non-increasing in a . An analogous argument, with the channel T modified to move mass from outcome “other” to outcome j instead of outcome i , shows that $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ is also non-increasing in b .

■

From Lemma C.2, for fixed ε , P_{nse} , and β , the maximum $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ is achieved at the smallest values of a and b , denoted by a_{min} and b_{min} . Recall that $a = P'_{syn}(i)$ and $b = P'_{syn}(j)$ where,

$$P'_{syn}(y) = (1 - \beta)\bar{P}'_{prv}(y) + \beta P_{nse}(y), \quad \forall y \in \mathcal{X} \quad (44)$$

Thus, the smallest a and b for fixed β and P_{nse} are given by,

$$a_{min} = \beta N_{min_1} \quad (45)$$

$$b_{min} = \frac{1 - \beta}{n} + \beta N_{min_2} \quad (46)$$

or in the reverse order:

$$a'_{min} = \beta N_{min_2} \quad (47)$$

$$b'_{min} = \frac{1 - \beta}{n} + \beta N_{min_1} \quad (48)$$

where N_{min_1} and N_{min_2} are the two smallest elements of P_{nse} . This simplifies the privacy constraint in Lemma C.1 to:

$$\max\{f(a_{min}, b_{min}, \varepsilon, \beta), f(a'_{min}, b'_{min}, \varepsilon, \beta)\} \leq \delta \quad (49)$$

Thus, for given ε, δ, m and P_{nse} , the optimum mixing parameter is given by,

$$\beta_{min}(P_{nse}) = \inf \{\beta \in [0, 1] : \max\{f(a_{min}, b_{min}, \varepsilon, \beta), f(a'_{min}, b'_{min}, \varepsilon, \beta)\} \leq \delta\} \quad (50)$$

where we have ignored the dependency of β on ε, δ and m since they are fixed for a given setting, and P_{nse} is the only variable that is yet to be optimized. Note that $\beta_{min}(P_{nse})$ depends on P_{nse} only through its two smallest elements. Thus, for a given P_{nse} , the minimum mixing coefficient is given by:

$$\beta_{min}(N_{min_1}, N_{min_2}) = \inf \{\beta \in [0, 1] : \max\{f(a_{min}, b_{min}, \varepsilon, \beta), f(a'_{min}, b'_{min}, \varepsilon, \beta)\} \leq \delta\} \quad (51)$$

where

$$f(a_{min}, b_{min}, \varepsilon, \beta) = \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} \left(1 - \beta(N_{min_1} + N_{min_2}) - \frac{1 - \beta}{n}\right)^{m - \lambda - \mu} \left[\left(\beta N_{min_1} + \frac{1 - \beta}{n}\right)^\lambda (\beta N_{min_2})^\mu - e^\varepsilon \left(\beta N_{min_2} + \frac{1 - \beta}{n}\right)^\mu (\beta N_{min_1})^\lambda \right]_+ \quad (52)$$

and

$$f(a'_{min}, b'_{min}, \varepsilon, \beta) = \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} \left(1 - \beta(N_{min_1} + N_{min_2}) - \frac{1 - \beta}{n}\right)^{m - \lambda - \mu} \left[\left(\beta N_{min_2} + \frac{1 - \beta}{n}\right)^\lambda (\beta N_{min_1})^\mu - e^\varepsilon \left(\beta N_{min_1} + \frac{1 - \beta}{n}\right)^\mu (\beta N_{min_2})^\lambda \right]_+ \quad (53)$$

Next, we show that $f(a_{min}, b_{min}, \varepsilon, \beta)$ and $f(a'_{min}, b'_{min}, \varepsilon, \beta)$ are decreasing in β . Thus, for any P_{nse} with a fixed N_{min_1} and N_{min_2} , any $\beta \geq \beta(N_{min_1}, N_{min_2})$ satisfies the privacy constraint in (2).

Lemma C.3. For any fixed N_{min_1}, N_{min_2} and $\varepsilon > 0$, $\max\{f(a_{min}, b_{min}, \varepsilon, \beta), f(a'_{min}, b'_{min}, \varepsilon, \beta)\}$ is decreasing in β .

Proof: From the monotonicity of $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ in a and b established in the proof of Lemma C.2, for any fixed $\beta, \varepsilon, N_{min_1}$ and N_{min_2} , we have the privacy constraint in (2) simplified to:

$$\sup_{D_{prv}, D'_{prv}} E_{e^\varepsilon}(P^{\otimes m}_{syn} \| P^{\otimes m}_{syn}) = \sup_{(a, b) \in \mathcal{F}(\beta, N_{min_1}, N_{min_2})} E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m}) = \max\{E_{e^\varepsilon}(P^{\otimes m}_* \| Q^{\otimes m}_*), E_{e^\varepsilon}(P^{\otimes m}_\# \| Q^{\otimes m}_\#)\} \leq \delta \quad (54)$$

where

$$P_* = \left[a_{\min} + \frac{1-\beta}{n}, b_{\min} - \frac{1-\beta}{n}, 1 - a_{\min} - b_{\min} \right] \quad (55)$$

$$Q_* = [a_{\min}, b_{\min}, 1 - a_{\min} - b_{\min}] \quad (56)$$

and

$$P_{\#} = \left[a'_{\min} + \frac{1-\beta}{n}, b'_{\min} - \frac{1-\beta}{n}, 1 - a'_{\min} - b'_{\min} \right] \quad (57)$$

$$Q_{\#} = [a'_{\min}, b'_{\min}, 1 - a'_{\min} - b'_{\min}] \quad (58)$$

with $a_{\min} = \beta N_{\min_1}$, $b_{\min} = \frac{1-\beta}{n} + \beta N_{\min_2}$ and $a'_{\min} = \beta N_{\min_2}$ and $b'_{\min} = \frac{1-\beta}{n} + \beta N_{\min_1}$. Note that $f(a_{\min}, b_{\min}, \varepsilon, \beta) = E_{e^\varepsilon}(P_*^{\otimes m} \| Q_*^{\otimes m})$ and $f(a'_{\min}, b'_{\min}, \varepsilon, \beta) = E_{e^\varepsilon}(P_{\#}^{\otimes m} \| Q_{\#}^{\otimes m})$. Moreover, $a_{\min}, a'_{\min}, b_{\min}, b'_{\min}$ are functions of β . Suppose for a given $N_{\min_1}, N_{\min_2}, \beta_0$ satisfies:⁶

$$\max\{f(a_{\min}(\beta_0), b_{\min}(\beta_0), \varepsilon, \beta_0), f(a'_{\min}(\beta_0), b'_{\min}(\beta_0), \varepsilon, \beta_0)\} \leq \delta \quad (59)$$

In other words, the privacy constraint holds for all neighboring datasets at $\beta = \beta_0$. For any $\beta > \beta_0$, we can write:

$$\begin{aligned} P_{\text{syn}, \beta} &= (1 - \beta) \bar{P}_{\text{prv}} + \beta P_{\text{nse}} \\ &= \frac{1 - \beta}{1 - \beta_0} [(1 - \beta_0) \bar{P}_{\text{prv}} + \beta_0 P_{\text{nse}}] + \left(1 - \frac{1 - \beta}{1 - \beta_0}\right) P_{\text{nse}} \\ &= \alpha P_{\text{syn}, \beta_0} + (1 - \alpha) P_{\text{nse}} \end{aligned} \quad (60)$$

where $\alpha = \frac{1-\beta}{1-\beta_0} \in [0, 1]$ and $P_{\text{syn}, \beta_0} = (1 - \beta_0) \bar{P}_{\text{prv}} + \beta_0 P_{\text{nse}}$. Similarly, for any neighboring dataset:

$$P'_{\text{syn}, \beta} = \alpha P'_{\text{syn}, \beta_0} + (1 - \alpha) P_{\text{nse}}. \quad (61)$$

That is, both $P_{\text{syn}, \beta}$ and $P'_{\text{syn}, \beta}$ are obtained by applying the same stochastic channel T to P_{syn, β_0} and P'_{syn, β_0} , respectively, where T takes the synthetic distribution as input and outputs a sample from the synthetic distribution with mixing parameter β_0 with probability α and a sample from P_{nse} with probability $1 - \alpha$. Since T is applied independently to each of the m samples, by the data processing inequality for the hockey-stick divergence:

$$E_{e^\varepsilon}(P_{\text{syn}, \beta}^{\otimes m} \| P'_{\text{syn}, \beta}{}^{\otimes m}) \leq E_{e^\varepsilon}(P_{\text{syn}, \beta_0}^{\otimes m} \| P'_{\text{syn}, \beta_0}{}^{\otimes m}) \quad (62)$$

Since (62) holds for every neighboring pair $(D_{\text{prv}}, D'_{\text{prv}})$, and the channel T does not depend on the choice of neighboring pair, we can take the supremum over all neighboring pairs on both sides:

$$\sup_{D_{\text{prv}}, D'_{\text{prv}}} E_{e^\varepsilon}(P_{\text{syn}, \beta}^{\otimes m} \| P'_{\text{syn}, \beta}{}^{\otimes m}) \leq \sup_{D_{\text{prv}}, D'_{\text{prv}}} E_{e^\varepsilon}(P_{\text{syn}, \beta_0}^{\otimes m} \| P'_{\text{syn}, \beta_0}{}^{\otimes m}) \leq \delta. \quad (63)$$

where the last inequality follows from the fact that privacy is guaranteed at $\beta = \beta_0$. From the monotonicity of $E_{e^\varepsilon}(P^{\otimes m} \| Q^{\otimes m})$ in a and b established in (43), the supremum over all neighboring pairs at any fixed β is achieved at $(a_{\min}, b_{\min})$ or $(a'_{\min}, b'_{\min})$ for a given P_{nse} . Therefore,

$$\begin{aligned} \max\{f(a_{\min}(\beta), b_{\min}(\beta), \varepsilon, \beta), f(a'_{\min}(\beta), b'_{\min}(\beta), \varepsilon, \beta)\} &\leq \max\{f(a_{\min}(\beta_0), b_{\min}(\beta_0), \varepsilon, \beta_0), f(a'_{\min}(\beta_0), b'_{\min}(\beta_0), \varepsilon, \beta_0)\} \\ &\leq \delta, \end{aligned} \quad (64)$$

establishing that $\max\{f(a_{\min}, b_{\min}, \varepsilon, \beta), f(a'_{\min}, b'_{\min}, \varepsilon, \beta)\}$ is decreasing in β . ■

With Lemma C.3 and the characterization of $\beta_{\min}(N_{\min_1}, N_{\min_2})$ in (51), we claim that for any fixed $N_{\min_1}, N_{\min_2}$ and ε , any $\beta \geq \beta(N_{\min_1}, N_{\min_2})$ satisfies the privacy constraint in (2). To simplify the process of computing $\beta_{\min}(N_{\min_1}, N_{\min_2})$ for given $N_{\min_1}, N_{\min_2}$ from (51), we next show that $\max\{f(a_{\min}(\beta), b_{\min}(\beta), \varepsilon, \beta), f(a'_{\min}(\beta), b'_{\min}(\beta), \varepsilon, \beta)\}$ is continuous in β (see Lemma C.4). Then, together with the fact that $\max\{f(a_{\min}(\beta), b_{\min}(\beta), \varepsilon, \beta), f(a'_{\min}(\beta), b'_{\min}(\beta), \varepsilon, \beta)\}$ is decreasing in β (Lemma C.3), we find $\beta_{\min}(N_{\min_1}, N_{\min_2})$ for any given $N_{\min_1}, N_{\min_2}$ by solving:

$$\max\{f(a_{\min}(\beta), b_{\min}(\beta), \varepsilon, \beta), f(a'_{\min}(\beta), b'_{\min}(\beta), \varepsilon, \beta)\} = \delta \quad (65)$$

⁶Since $a_{\min}, a'_{\min}, b_{\min}, b'_{\min}$ are functions of β , we make it explicit by writing $f(a_{\min}, b_{\min}, \varepsilon, \beta) = f(a_{\min}(\beta), b_{\min}(\beta), \varepsilon, \beta)$ and $f(a'_{\min}, b'_{\min}, \varepsilon, \beta) = f(a'_{\min}(\beta), b'_{\min}(\beta), \varepsilon, \beta)$.

Lemma C.4. For any fixed N_{min_1}, N_{min_2} and ε , $\max\{f(a_{min}(\beta), b_{min}(\beta), \varepsilon, \beta), f(a'_{min}(\beta), b'_{min}(\beta), \varepsilon, \beta)\}$ is continuous in β .

Proof: Consider:

$$f(a_{min}(\beta), b_{min}(\beta), \varepsilon, \beta) = \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} \left(1 - \beta(N_{min_1} + N_{min_2}) - \frac{1 - \beta}{n}\right)^{m - \lambda - \mu} \left[\left(\beta N_{min_1} + \frac{1 - \beta}{n}\right)^\lambda (\beta N_{min_2})^\mu - e^\varepsilon \left(\beta N_{min_2} + \frac{1 - \beta}{n}\right)^\mu (\beta N_{min_1})^\lambda \right]_+ \quad (66)$$

Each term in the summation is indexed by a fixed (λ, μ) with $\lambda, \mu \geq 0$ and $\lambda + \mu \leq m$, and takes the form

$$g_{\lambda, \mu}(\beta) = \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} c(\beta)^{m - \lambda - \mu} [r_1(\beta)^\lambda q_2(\beta)^\mu - e^\varepsilon r_2(\beta)^\mu q_1(\beta)^\lambda]_+ \quad (67)$$

where $c(\beta) = 1 - \beta(N_{min_1} + N_{min_2}) - \frac{1 - \beta}{n}$, $q_i(\beta) = \beta N_{min_i}$, and $r_i(\beta) = \beta N_{min_i} + \frac{1 - \beta}{n}$. Each of $c(\beta)$, $q_i(\beta)$, and $r_i(\beta)$ is continuous in β on $[0, 1]$, and therefore $h(\beta) \triangleq r_1(\beta)^\lambda q_2(\beta)^\mu - e^\varepsilon r_2(\beta)^\mu q_1(\beta)^\lambda$ is also continuous in β . Since the function $x \mapsto [x]_+ = \max\{0, x\}$ is continuous, the composition $[h(\beta)]_+$ is continuous. It follows that each $g_{\lambda, \mu}(\beta)$ is a product of continuous functions and hence continuous. Since the summation is over a finite set of (λ, μ) pairs that does not depend on β , the function

$$f(a_{min}(\beta), b_{min}(\beta), \varepsilon, \beta) = \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} g_{\lambda, \mu}(\beta) \quad (68)$$

is a finite sum of continuous functions, and is therefore continuous in β . Similarly, $f(a'_{min}(\beta), b'_{min}(\beta), \varepsilon, \beta)$ is also continuous in β . Thus, the maximum of the two functions is also continuous in β . ■

Lemma C.5. $\beta_{min}(N_{min_1}, N_{min_2})$ as defined in (51) is non-increasing in both N_{min_1} and N_{min_2} .

Proof: Recall the two worst-case pairs:

$$(a_{min}, b_{min}) = \left(\beta N_{min_1}, \frac{1 - \beta}{n} + \beta N_{min_2}\right), \quad (69)$$

$$(a'_{min}, b'_{min}) = \left(\beta N_{min_2}, \frac{1 - \beta}{n} + \beta N_{min_1}\right). \quad (70)$$

All $a_{min}, b_{min}, a'_{min}, b'_{min}$ are linear and non-decreasing in N_{min_1} and N_{min_2} as $\beta \in [0, 1]$. Since $f(a, b, \varepsilon, \beta)$ is non-increasing in a and b (Lemma C.2), it follows that for any fixed β , $f(a_{min}, b_{min}, \varepsilon, \beta)$ and $f(a'_{min}, b'_{min}, \varepsilon, \beta)$ are each non-increasing in N_{min_1} and N_{min_2} . Therefore their maximum,

$$g(\beta, N_{min_1}, N_{min_2}) := \max\{f(a_{min}, b_{min}, \varepsilon, \beta), f(a'_{min}, b'_{min}, \varepsilon, \beta)\}, \quad (71)$$

is also non-increasing in N_{min_1} and N_{min_2} for any fixed β .

Now fix any $N_{min_1} \leq N'_{min_1}$ (the argument for N_{min_2} is identical). Let $\beta^* = \beta_{min}(N_{min_1}, N_{min_2})$ in (51), so that by definition:

$$g(\beta^*, N_{min_1}, N_{min_2}) \leq \delta. \quad (72)$$

Since g is non-increasing in N_{min_1} :

$$g(\beta^*, N'_{min_1}, N_{min_2}) \leq g(\beta^*, N_{min_1}, N_{min_2}) \leq \delta, \quad (73)$$

Therefore, β^* is in the feasible range in (51):

$$\beta^* \in \{\beta \in [0, 1] : g(\beta, N'_{min_1}, N_{min_2}) \leq \delta\} \quad (74)$$

Since $\beta_{min}(N'_{min_1}, N_{min_2})$ is the infimum over all feasible β :

$$\beta_{min}(N'_{min_1}, N_{min_2}) \leq \beta^* = \beta_{min}(N_{min_1}, N_{min_2}). \quad (75)$$

Hence $\beta_{min}(N_{min_1}, N_{min_2})$ is non-increasing in N_{min_1} , and by the same argument, in N_{min_2} . ■

This completes the proof of Theorem 4.1.

D. Details from Section 4

D.1. Proof of Asymptotic Optimality

The optimization in (8) replaces $\beta_{\min}(N_{\min_1}, N_{\min_2})$ by the more conservative quantity $\beta_{\min}(N_{\min_1}, N_{\min_1})$. Since $N_{\min_2} \geq N_{\min_1}$ and $\beta_{\min}(N_{\min_1}, N_{\min_2})$ is non-increasing in its arguments (Lemma C.5), this yields an upper bound on the original objective. Moreover, the simplex constraint $N_{\min_1} + (d-1)N_{\min_2} \leq 1$ implies:

$$0 \leq N_{\min_2} - N_{\min_1} \leq \frac{1}{d-1}. \quad (76)$$

Lemma D.1. *For any fixed $N_{\min_1} \in [0, 1/d]$, $\beta_{\min}(N_{\min_1}, N_{\min_2})$ is continuous in $N_{\min_2}$. Consequently,*

$$\sup_{P_{\text{nse}} \in \Delta_d} |\beta_{\min}(N_{\min_1}, N_{\min_2}) - \beta_{\min}(N_{\min_1}, N_{\min_1})| \rightarrow 0 \quad \text{as } d \rightarrow \infty. \quad (77)$$

Moreover, the optimal value of the upper-bound problem in (8) converges to the optimal value of the original problem. Any optimizer of the upper-bound problem is therefore asymptotically optimal for the original problem.

Proof. Assume that $\beta_{\min}(N_{\min_1}, N_{\min_2})$ is continuous in $N_{\min_2}$. The proof is provided at the end.

Let $N_1(P_{\text{nse}}) := N_{\min_1}$ and $N_2(P_{\text{nse}}) := N_{\min_2}$, where $N_1(P_{\text{nse}}) \leq N_2(P_{\text{nse}})$ are the two smallest coordinates of P_{nse} . Define

$$\Psi(P_{\text{nse}}) := \max_{\bar{P}_{\text{prv}} \in B_{\gamma, \bar{P}_{\text{pub}}}} TV(\bar{P}_{\text{prv}}, P_{\text{nse}}). \quad (78)$$

After optimizing over β , the objective in the left-hand side of (8) can be written as

$$F_d(P_{\text{nse}}) = \beta_{\min}(N_1(P_{\text{nse}}), N_2(P_{\text{nse}})) \Psi(P_{\text{nse}}), \quad (79)$$

while the upper-bound objective in the right-hand side of (8) is

$$\tilde{F}_d(P_{\text{nse}}) = \beta_{\min}(N_1(P_{\text{nse}}), N_1(P_{\text{nse}})) \Psi(P_{\text{nse}}). \quad (80)$$

Since $N_2(P_{\text{nse}}) \geq N_1(P_{\text{nse}})$, and since $\beta_{\min}$ is non-increasing in each of its arguments, we have

$$\beta_{\min}(N_1(P_{\text{nse}}), N_2(P_{\text{nse}})) \leq \beta_{\min}(N_1(P_{\text{nse}}), N_1(P_{\text{nse}})). \quad (81)$$

Therefore,

$$F_d(P_{\text{nse}}) \leq \tilde{F}_d(P_{\text{nse}}) \quad \forall P_{\text{nse}} \in \Delta_d. \quad (82)$$

Thus,

$$V_d := \inf_{P_{\text{nse}} \in \Delta_d} F_d(P_{\text{nse}}) \leq \inf_{P_{\text{nse}} \in \Delta_d} \tilde{F}_d(P_{\text{nse}}) =: \tilde{V}_d, \quad (83)$$

so the right-hand side of (8) is indeed an upper bound on the original objective.

Next, we bound the gap between the two objectives. For any $P_{\text{nse}} \in \Delta_d$,

$$0 \leq \tilde{F}_d(P_{\text{nse}}) - F_d(P_{\text{nse}}) = [\beta_{\min}(N_1(P_{\text{nse}}), N_1(P_{\text{nse}})) - \beta_{\min}(N_1(P_{\text{nse}}), N_2(P_{\text{nse}}))] \Psi(P_{\text{nse}}). \quad (84)$$

Since total variation distance is at most 1, we have

$$0 \leq \Psi(P_{\text{nse}}) \leq 1. \quad (85)$$

Hence,

$$0 \leq \tilde{F}_d(P_{\text{nse}}) - F_d(P_{\text{nse}}) \leq \beta_{\min}(N_1(P_{\text{nse}}), N_1(P_{\text{nse}})) - \beta_{\min}(N_1(P_{\text{nse}}), N_2(P_{\text{nse}})). \quad (86)$$

We now use the simplex constraint. Since $P_{\text{nse}} \in \Delta_d$, its coordinates sum to one. Because N_1 and N_2 are the smallest and second-smallest coordinates, all coordinates except the smallest one are at least N_2 . Therefore, for any $P_{\text{nse}} \in \Delta_d$,

$$1 = \sum_{i=1}^d P_{\text{nse}}(i) \geq N_1 + (d-1)N_2. \quad (87)$$

Thus,

$$N_2 \leq \frac{1 - N_1}{d-1}. \quad (88)$$

Since $N_2 \geq N_1$, we obtain

$$0 \leq N_2 - N_1 \leq \frac{1 - N_1}{d-1} - N_1 = \frac{1 - dN_1}{d-1} \leq \frac{1}{d-1}. \quad (89)$$

Therefore,

$$0 \leq N_2 - N_1 \leq \frac{1}{d-1} \quad (90)$$

Assume that $\beta_{\min}(z_1, z_2)$ is uniformly continuous in its second argument over the feasible region

$$\mathcal{A}_d = \{(z_1, z_2) : 0 \leq z_1 \leq z_2, z_1 + (d-1)z_2 \leq 1\}. \quad (91)$$

Then, since

$$|N_2(P_{\text{nse}}) - N_1(P_{\text{nse}})| \leq \frac{1}{d-1}, \quad \forall P_{\text{nse}} \in \Delta_d \quad (92)$$

we have

$$\sup_{P_{\text{nse}} \in \Delta_d} |\beta_{\min}(N_1(P_{\text{nse}}), N_2(P_{\text{nse}})) - \beta_{\min}(N_1(P_{\text{nse}}), N_1(P_{\text{nse}}))| \rightarrow 0 \quad (93)$$

as $d \rightarrow \infty$. Consequently,

$$\sup_{P_{\text{nse}} \in \Delta_d} |\tilde{F}_d(P_{\text{nse}}) - F_d(P_{\text{nse}})| \rightarrow 0. \quad (94)$$

It remains to show convergence of the optimal values. Since $F_d(P_{\text{nse}}) \leq \tilde{F}_d(P_{\text{nse}})$ pointwise, we have

$$V_d \leq \tilde{V}_d. \quad (95)$$

Let $P_d^* \in \arg \min_{P_{\text{nse}} \in \Delta_d} F_d(P_{\text{nse}})$. Then

$$\tilde{V}_d = \inf_{P_{\text{nse}} \in \Delta_d} \tilde{F}_d(P_{\text{nse}}) \leq \tilde{F}_d(P_d^*). \quad (96)$$

Therefore,

$$\tilde{V}_d \leq F_d(P_d^*) + \sup_{P_{\text{nse}} \in \Delta_d} |\tilde{F}_d(P_{\text{nse}}) - F_d(P_{\text{nse}})|. \quad (97)$$

Since $F_d(P_d^*) = V_d$, we get

$$0 \leq \tilde{V}_d - V_d \leq \sup_{P_{\text{nse}} \in \Delta_d} |\tilde{F}_d(P_{\text{nse}}) - F_d(P_{\text{nse}})|. \quad (98)$$

Taking $d \rightarrow \infty$, the right-hand side converges to zero. Hence,

$$\tilde{V}_d - V_d \rightarrow 0. \quad (99)$$

Thus, the upper-bound problem in (8) has the same asymptotic optimal value as the original problem. Moreover, if

$$\tilde{P}_d^* \in \arg \min_{P_{\text{nse}} \in \Delta_d} \tilde{F}_d(P_{\text{nse}}), \quad (100)$$

then

$$F_d(\tilde{P}_d^*) \leq \tilde{F}_d(\tilde{P}_d^*) = \tilde{V}_d. \quad (101)$$

Therefore,

$$0 \leq F_d(\tilde{P}_d^*) - V_d \leq \tilde{V}_d - V_d \rightarrow 0. \quad (102)$$

Hence, any optimizer of the upper-bound problem is asymptotically optimal for the original problem.

Now, the only remaining component of the proof is the continuity of $\beta_{\min}(N_{\min_1}, N_{\min_2})$ in $N_{\min_2}$.

Continuity of $\beta_{\min}(N_{\min_1}, N_{\min_2})$ in $N_{\min_2}$.

Fix $\varepsilon > 0$, $\delta \in (0, 1)$, the private dataset size n , the number of released samples m , and $N_{\min_1}$. For notational simplicity, write

$$z_1 := N_{\min_1}, \quad z_2 := N_{\min_2}. \quad (103)$$

Recall from Theorem 4.1 that $\beta_{\min}(z_1, z_2)$ is defined as

$$\beta_{\min}(z_1, z_2) = \inf \{ \beta \in [0, 1] : G(\beta, z_1, z_2) = \delta \}, \quad (104)$$

where

$$G(\beta, z_1, z_2) := \max \{ h_1(\beta, \varepsilon, n, m, z_1, z_2), h_2(\beta, \varepsilon, n, m, z_1, z_2) \}. \quad (105)$$

Here h_1 and h_2 are the expressions from Theorem 4.1.

We first observe that G is continuous in (β, z_1, z_2) . Indeed, h_1 is a finite sum of terms of the form:

$$\begin{aligned} & h_1(\beta, \varepsilon, n, m, z_1, z_2) \\ &= \sum_{\substack{\lambda, \mu \geq 0 \\ \lambda + \mu \leq m}} \frac{m!}{\lambda! \mu! (m - \lambda - \mu)!} \left(1 - \beta(z_1 + z_2) - \frac{1 - \beta}{n} \right)^{m - \lambda - \mu} \left[\left(\beta z_1 + \frac{1 - \beta}{n} \right)^\lambda (\beta z_2)^\mu - e^\varepsilon \left(\beta z_2 + \frac{1 - \beta}{n} \right)^\mu (\beta z_1)^\lambda \right]_+ \end{aligned} \quad (106)$$

Each factor inside the summation is continuous in (β, z_1, z_2) , and the map $x \mapsto [x]_+ = \max\{x, 0\}$ is continuous. Therefore h_1 is continuous. The same argument applies to h_2 , and since the maximum of two continuous functions is continuous, G is continuous.

Now fix z_1 , and consider $z_2 \mapsto \beta_{\min}(z_1, z_2)$. Let

$$z_2^{(k)} \rightarrow z_2. \quad (107)$$

We show that

$$\beta_{\min}(z_1, z_2^{(k)}) \rightarrow \beta_{\min}(z_1, z_2). \quad (108)$$

Let

$$\beta^* := \beta_{\min}(z_1, z_2). \quad (109)$$

Then, we have,

$$G(\beta^*, z_1, z_2) = \delta, \quad (110)$$

Claim D.2. For every $\eta > 0$,

$$G(\beta^* - \eta, z_1, z_2) > \delta, \quad G(\beta^* + \eta, z_1, z_2) < \delta. \quad (111)$$

Equivalently, the privacy curve crosses the level δ at β^* , rather than remaining flat at δ .

The proof of Claim D.2 is given at the end. Using Claim D.2, we prove upper and lower semicontinuity.

First, fix any $\eta > 0$. By Claim D.2,

$$G(\beta^* + \eta, z_1, z_2) < \delta. \quad (112)$$

Since G is continuous in z_2 , for all sufficiently large k ,

$$G(\beta^* + \eta, z_1, z_2^{(k)}) < \delta. \quad (113)$$

Therefore,

$$\beta_{\min}(z_1, z_2^{(k)}) \leq \beta^* + \eta \quad (114)$$

for all sufficiently large k . Taking the limsup gives

$$\limsup_{k \rightarrow \infty} \beta_{\min}(z_1, z_2^{(k)}) \leq \beta^* + \eta. \quad (115)$$

Since $\eta > 0$ was arbitrary,

$$\limsup_{k \rightarrow \infty} \beta_{\min}(z_1, z_2^{(k)}) \leq \beta^*. \quad (116)$$

Next, again fix any $\eta > 0$. By Claim D.2,

$$G(\beta^* - \eta, z_1, z_2) > \delta. \quad (117)$$

By continuity of G in z_2 , for all sufficiently large k ,

$$G(\beta^* - \eta, z_1, z_2^{(k)}) > \delta. \quad (118)$$

Thus $\beta^* - \eta$ is not feasible for the problem defining $\beta_{\min}(z_1, z_2^{(k)})$. Since G is non-increasing in β , no value

$$\beta \leq \beta^* - \eta$$

is feasible either. Therefore,

$$\beta_{\min}(z_1, z_2^{(k)}) \geq \beta^* - \eta$$

for all sufficiently large k . Taking the liminf gives

$$\liminf_{k \rightarrow \infty} \beta_{\min}(z_1, z_2^{(k)}) \geq \beta^* - \eta.$$

Since $\eta > 0$ was arbitrary,

$$\liminf_{k \rightarrow \infty} \beta_{\min}(z_1, z_2^{(k)}) \geq \beta^*.$$

Combining the two inequalities,

$$\limsup_{k \rightarrow \infty} \beta_{\min}(z_1, z_2^{(k)}) \leq \beta^* \leq \liminf_{k \rightarrow \infty} \beta_{\min}(z_1, z_2^{(k)}).$$

Hence,

$$\beta_{\min}(z_1, z_2^{(k)}) \rightarrow \beta^* = \beta_{\min}(z_1, z_2).$$

Therefore, $\beta_{\min}(N_{\min_1}, N_{\min_2})$ is continuous in $N_{\min_2}$. □

Proof of Claim D.2

Proof. The first inequality in (111) is straightforward from the definition of β^* . For the second inequality, we show that for every $\eta > 0$ such that $\beta^* + \eta \leq 1$,

$$G(\beta^* + \eta, z_1, z_2) < \delta.$$

Let $\beta_0 := \beta^*$ and $\beta_1 := \beta^* + \eta$. Since $\eta > 0$, we have $\beta_1 > \beta_0$. Define $\alpha := \frac{1-\beta_1}{1-\beta_0}$. Then $0 \leq \alpha < 1$. Now consider any neighboring pair of private histograms $\bar{P}_{\text{prv}}$ and $\bar{P}'_{\text{prv}}$. For a fixed noise distribution P_{nse} , define

$$P_{\text{syn},\beta} = (1 - \beta)\bar{P}_{\text{prv}} + \beta P_{\text{nse}},$$

and

$$P'_{\text{syn},\beta} = (1 - \beta)\bar{P}'_{\text{prv}} + \beta P_{\text{nse}}.$$

Then

$$P_{\text{syn},\beta_1} = \alpha P_{\text{syn},\beta_0} + (1 - \alpha)P_{\text{nse}},$$

and similarly

$$P'_{\text{syn},\beta_1} = \alpha P'_{\text{syn},\beta_0} + (1 - \alpha)P_{\text{nse}}.$$

For compactness, write

$$P_0 := P_{\text{syn},\beta_0}, \quad Q_0 := P'_{\text{syn},\beta_0}, \quad R := P_{\text{nse}}.$$

Then

$$P_1 := P_{\text{syn},\beta_1} = \alpha P_0 + (1 - \alpha)R,$$

and

$$Q_1 := P'_{\text{syn},\beta_1} = \alpha Q_0 + (1 - \alpha)R.$$

We now compare the m -fold product distributions. Since

$$P_1^{\otimes m} = (\alpha P_0 + (1 - \alpha)R)^{\otimes m},$$

we can expand⁷

$$P_1^{\otimes m} = \sum_{S \subseteq [m]} \alpha^{|S|} (1 - \alpha)^{m-|S|} P_0^{\otimes |S|} \otimes R^{\otimes (m-|S|)}.$$

Similarly,

$$Q_1^{\otimes m} = \sum_{S \subseteq [m]} \alpha^{|S|} (1 - \alpha)^{m-|S|} Q_0^{\otimes |S|} \otimes R^{\otimes (m-|S|)}.$$

Using convexity of the hockey-stick divergence in the pair of distributions, we obtain

$$E_\gamma(P_1^{\otimes m} \| Q_1^{\otimes m}) \leq \sum_{S \subseteq [m]} \alpha^{|S|} (1 - \alpha)^{m-|S|} E_\gamma \left(P_0^{\otimes |S|} \otimes R^{\otimes (m-|S|)} \left\| Q_0^{\otimes |S|} \otimes R^{\otimes (m-|S|)} \right\| \right).$$

Since the same distribution $R^{\otimes (m-|S|)}$ appears under both hypotheses on the coordinates in S^c , those coordinates contain no information for distinguishing the two distributions. By data processing, or equivalently by marginalizing out the coordinates in S^c ,

$$E_\gamma \left(P_0^{\otimes S} \otimes R^{\otimes S^c} \left\| Q_0^{\otimes S} \otimes R^{\otimes S^c} \right\| \right) = E_\gamma \left(P_0^{\otimes |S|} \left\| Q_0^{\otimes |S|} \right\| \right).$$

⁷Sampling one coordinate from P_1 is equivalent to first drawing a Bernoulli variable with success probability α . With probability α , the coordinate is sampled from P_0 , and with probability $1 - \alpha$, it is sampled from R . For m independent samples, let $S \subseteq [m]$ denote the set of coordinates sampled from P_0 . Then the coordinates in S^c are sampled from R . The probability of this choice of S is $\alpha^{|S|} (1 - \alpha)^{m-|S|}$, and the corresponding product distribution is $P_0^{\otimes |S|} \otimes R^{\otimes |S^c|}$. Summing over all subsets $S \subseteq [m]$ gives the stated expansion.

Therefore,

$$E_\gamma(P_1^{\otimes m} \| Q_1^{\otimes m}) \leq \sum_{S \subseteq [m]} \alpha^{|S|} (1 - \alpha)^{m-|S|} E_\gamma(P_0^{\otimes |S|} \| Q_0^{\otimes |S|}).$$

Grouping subsets by cardinality $s = |S|$, this becomes

$$E_\gamma(P_1^{\otimes m} \| Q_1^{\otimes m}) \leq \sum_{s=0}^m \binom{m}{s} \alpha^s (1 - \alpha)^{m-s} E_\gamma(P_0^{\otimes s} \| Q_0^{\otimes s}).$$

For $s = 0$, the two distributions are identical, so

$$E_\gamma(P_0^{\otimes 0} \| Q_0^{\otimes 0}) = 0.$$

For every $1 \leq s \leq m$, marginalizing an m -sample observation down to its first s coordinates is a stochastic channel. Hence, by data processing,

$$E_\gamma(P_0^{\otimes s} \| Q_0^{\otimes s}) \leq E_\gamma(P_0^{\otimes m} \| Q_0^{\otimes m}).$$

Thus,

$$E_\gamma(P_1^{\otimes m} \| Q_1^{\otimes m}) \leq \sum_{s=1}^m \binom{m}{s} \alpha^s (1 - \alpha)^{m-s} E_\gamma(P_0^{\otimes m} \| Q_0^{\otimes m}).$$

Since

$$\sum_{s=1}^m \binom{m}{s} \alpha^s (1 - \alpha)^{m-s} = 1 - (1 - \alpha)^m,$$

we get

$$E_\gamma(P_1^{\otimes m} \| Q_1^{\otimes m}) \leq (1 - (1 - \alpha)^m) E_\gamma(P_0^{\otimes m} \| Q_0^{\otimes m}).$$

Now $\gamma = e^\epsilon$. Therefore,

$$E_{e^\epsilon}(P_{\text{syn}, \beta_1}^{\otimes m} \| P_{\text{syn}, \beta_1}'^{\otimes m}) \leq (1 - (1 - \alpha)^m) E_{e^\epsilon}(P_{\text{syn}, \beta_0}^{\otimes m} \| P_{\text{syn}, \beta_0}'^{\otimes m}).$$

This inequality holds for every neighboring pair $(D_{\text{prv}}, D_{\text{prv}}')$. Taking the supremum over neighboring pairs gives

$$G(\beta_1, z_1, z_2) \leq (1 - (1 - \alpha)^m) G(\beta_0, z_1, z_2).$$

Using $\beta_0 = \beta^*$ and $G(\beta^*, z_1, z_2) = \delta$ we obtain,

$$G(\beta^* + \eta, z_1, z_2) \leq (1 - (1 - \alpha)^m) \delta.$$

Since $0 \leq \alpha < 1$ we have $0 < 1 - (1 - \alpha)^m < 1$ for $m \geq 1$ and $\alpha > 0$. Hence,

$$G(\beta^* + \eta, z_1, z_2) < \delta.$$

If $\alpha = 0$, then $\beta_1 = 1$, and

$$P_{\text{syn}, \beta_1} = P_{\text{syn}, \beta_1}' = P_{\text{nse}}.$$

Therefore,

$$G(1, z_1, z_2) = 0 < \delta.$$

Thus, in all cases,

$$G(\beta^* + \eta, z_1, z_2) < \delta$$

□

D.2. Interpretation of Remark 4.5:

To analyze the minmax TV distance in the LHS of (10), we split it into two cases.

$$\min_{P_{nse} \in \Delta_d} \underbrace{\min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn})}_{f(P_{nse})} = \min \left\{ \min_{P_{nse} \in \mathcal{S}} f(P_{nse}), \min_{P_{nse} \in \Delta_d \setminus \mathcal{S}} f(P_{nse}) \right\} \quad (119)$$

where $\mathcal{S} = \{P \in \Delta_d : \sum_i \bar{P}_{pub}^{[i]} \leq P^{[i]} \bar{P}_{pub}^{[i]} \geq \gamma\}$, with $P^{[i]}$ denoting the i th element of P in the same order as $\bar{P}_{pub}$. To interpret the two regimes for P_{nse} in (119), consider the simplified version of the LHS of (119) in (5). For a fixed P_{nse} , the inner max chooses the worst-case $\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}$ (i.e., farthest from P_{nse}). If $P_{nse} \in \mathcal{S}$, this worst-case $\bar{P}_{prv}$ lies on the same line through $\bar{P}_{pub}$ and P_{nse} , giving $\text{TV}(P_{nse}, \bar{P}_{pub}) + \gamma$ for the inner max (Fig. 6: blue). If $P_{nse} \in \Delta_d \setminus \mathcal{S}$, the worst-case $\bar{P}_{prv}$ is off that line, so the inner maximum is strictly smaller than $\text{TV}(P_{nse}, \bar{P}_{pub}) + \gamma$ (Fig. 6: red). These two cases require different analyses, which is provided in App. F. This results in the two-case structure in Remark 4.5.

The **Key insight from Remark 4.5** is that the approximation to the optimal noise distribution $\hat{P}_{nse}^*$ is always a floor-raised version of $\bar{P}_{pub}$, where both $\text{MIFR}(\bar{P}_{pub}, t_1)$ and $\text{WF}(\bar{P}_{pub}, t_2)$ raise small elements to their respective thresholds and redistribute probability mass from larger elements. This design balances two objectives: since $\bar{P}_{prv}$ and $\bar{P}_{pub}$ come from the same domain with typically small $\text{TV}(\bar{P}_{pub}, \bar{P}_{prv})$, choosing P_{nse} close to $\bar{P}_{pub}$ is beneficial. However, many elements in $\bar{P}_{pub}$ can be small (even zero), which would add minimal noise to corresponding $\bar{P}_{prv}$ elements and require a larger mixing weight β to maintain privacy, pushing P_{syn} too close to $\bar{P}_{pub}$ (not to $\bar{P}_{prv}$). The optimal P_{nse}^* resolves this tradeoff by remaining close to $\bar{P}_{pub}$ while floor-raising smaller probabilities to guarantee a sufficient $p = \min_i P_{nse}^{[i]}$, which allows a smaller mixing weight β and yields a P_{syn} that is closer to $\bar{P}_{prv}$.

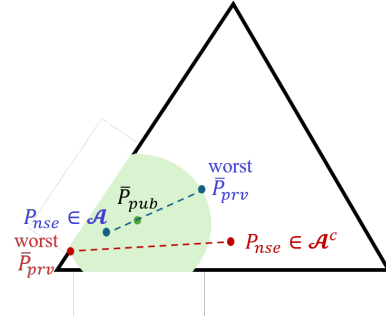


Figure 6. The two cases in Remark 4.5 correspond to two sets of potential noise distributions: 1) a worst-case $\bar{P}_{prv}$ in (3) that is on the same line as P_{nse} and $\bar{P}_{pub}$; 2) a worst-case $\bar{P}_{prv}$ that is not on the same line.

E. Proof of Theorem 4.2

Theorem 4.2 restated: Let $\gamma \in (0, 1]$ be a constant and let $\bar{P}_{pub} \in \Delta_d$ be a public distribution. Fix the privacy parameters $\varepsilon > 0$, $\delta \in (0, 1)$ and the required number of synthetic samples $m \geq 1$. Then,

$$\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) = \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) f(t) \quad (120)$$

where $f(t)$ is the solution to the following linear program:

$$f(t) = \min_{P_{nse} \in \Delta_d, r \in \mathbb{R}} r, \quad \text{s.t.} \quad r + P_{nse}(S) \geq \min\{\bar{P}_{pub}(S) + \gamma, 1\}, \quad \forall S \subseteq [d], \quad \mathbf{1}_d^T P_{nse} = 1, \quad P_{nse}^{[i]} \geq t, \quad \forall i \quad (121)$$

The corresponding optimum P_{nse}^* is the minimizing P_{nse} of (121) with $t = t^*$ where $t^* = \arg \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) f(t)$. The optimum β^* is given by $\beta^* = \beta_{\min}(t^*, t^*)$.

Proof: We begin the proof by re-writing the LHS of (120) as follows:

$$\begin{aligned} & \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \\ &= \min_{0 \leq t \leq \frac{1}{d}} \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} = t}} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{\min}(t, t)}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \end{aligned} \quad (122)$$

$$= \min_{0 \leq t \leq \frac{1}{d}} \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} = t}} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{\min}(t, t)}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \beta \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (123)$$

$$= \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} = t}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (124)$$

$$= \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} \geq t}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (125)$$

where (125) comes from the fact that $\beta_{\min}(t, t)$ is decreasing in t (see Lemma C.5). Next, we consider the inner maximization in (125).

Lemma E.1. For any fixed $P_{nse} \in \Delta_d$,

$$\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) = \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} [\min \{ \bar{P}_{pub}(S) + \gamma, 1 \} - P_{nse}(S)] \quad (126)$$

where $\bar{P}_{pub}(S) = \sum_{i \in S} \bar{P}_{pub}^{[i]}$ and $P_{nse}(S) = \sum_{i \in S} P_{nse}^{[i]}$. The cardinality of set S is denoted by $|S|$, and $[d] = \{1, \dots, d\}$.

The proof of Lemma E.1 is given in Sec. E.1. Using Lemma E.1 in (125), we have,

$$\begin{aligned} & \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \\ &= \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} \geq t}} \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} [\min \{ \bar{P}_{pub}(S) + \gamma, 1 \} - P_{nse}(S)] \end{aligned} \quad (127)$$

Now, fix $t \in [0, 1/d]$, and consider the inner minmax optimization in (127). Let:

$$f(t) = \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} \geq t}} \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} [\min \{ \bar{P}_{pub}(S) + \gamma, 1 \} - P_{nse}(S)] \quad (128)$$

Since $\bar{P}_{pub}$ and γ are given constants, for a given subset $S \subset [d]$ satisfying $1 \leq |S| \leq d-1$, define:

$$C_S = \min \{ \bar{P}_{pub}(S) + \gamma, 1 \} \quad (129)$$

Then, $f(t) = \min_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} \geq t}} \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} [C_S - P_{nse}(S)]$. Introduce an auxiliary scalar r representing an upper bound on the inner maximum. For any fixed $P_{nse} \in \Delta_d$,

$$r \geq \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} [C_S - P_{nse}(S)] \iff r \geq C_S - P_{nse}(S), \quad \forall S \subset [d], \quad 1 \leq |S| \leq d-1 \quad (130)$$

Therefore,

$$\begin{aligned} f(t) &= \min_{P_{nse} \in \Delta_d, r \in \mathbb{R}} r \\ &\text{s.t. } P_{nse}^{[i]} \geq t, \quad \forall i \\ &\quad \sum_{i=1}^d P_{nse}^{[i]} = 1 \\ &\quad P_{nse}(S) + r \geq C_S, \quad \forall S \subset [d], \quad 1 \leq |S| \leq d-1 \end{aligned} \quad (131)$$

Plugging (131) back in (127) completes the proof of Theorem 4.2. ■

E.1. Proof of Lemma E.1

Proof: We first prove that the RHS of (126) is an upper bound for the LHS of (126). Then we show equality via an achievability proof. By the definition of TV distance, we have,

$$\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) = \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \max_{S \subseteq [d]} (\bar{P}_{prv}(S) - P_{nse}(S)) \quad (132)$$

$$= \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \max_{\substack{S \subseteq [d] \\ 1 \leq |S| \leq d-1}} (\bar{P}_{prv}(S) - P_{nse}(S)) \quad (133)$$

$$= \max_{\substack{S \subseteq [d] \\ 1 \leq |S| \leq d-1}} \left(\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \bar{P}_{prv}(S) - P_{nse}(S) \right) \quad (134)$$

The set $\{S \subset [d] : 1 \leq |S| \leq d-1\}$ in (133) is due to the fact that $S = \emptyset$ and $S = [d]$ both result in $\bar{P}_{prv}(S) - P_{nse}(S) = 0$, which do not contribute to the maximum. The two maxima in (133) can be switched as the set of subsets in $\{S \subset [d] : 1 \leq |S| \leq d-1\}$ is finite. Next, consider $\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \bar{P}_{prv}(S)$. For any $\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}$, and any fixed $S \subset [d]$ satisfying $1 \leq |S| \leq d-1$,

$$\bar{P}_{prv}(S) - \bar{P}_{pub}(S) \leq \text{TV}(\bar{P}_{prv}, \bar{P}_{pub}) \leq \gamma \implies \bar{P}_{prv}(S) \leq \bar{P}_{pub}(S) + \gamma \quad (135)$$

Moreover, since $\bar{P}_{prv}(S) \leq 1$, we have,

$$\bar{P}_{prv}(S) \leq \min\{\bar{P}_{pub}(S) + \gamma, 1\} \quad (136)$$

Since this applies to any $\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}$, we have,

$$\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \bar{P}_{prv}(S) \leq \min\{\bar{P}_{pub}(S) + \gamma, 1\} \quad (137)$$

for any fixed $S \subset [d]$ satisfying $1 \leq |S| \leq d-1$. Next, we show this upper bound is achievable. Define:

$$\alpha = \min\{\gamma, 1 - \bar{P}_{pub}(S)\} \quad (138)$$

We will move α probability mass from S^c to S where $S^c = [d] \setminus S$ denotes the complement of S . This is feasible because:

$$\alpha \leq 1 - \bar{P}_{pub}(S) = \bar{P}_{pub}(S^c) \quad (139)$$

Thus, there's sufficient mass to be removed from S^c . At the same time, there's sufficient capacity in S to receive α probability mass as $\alpha \leq 1 - \bar{P}_{pub}(S)$. Now, construct a $\hat{P}_{prv}$ as follows (denote it by $\hat{P}_{prv}$ to avoid confusion):

$$\hat{P}_{prv}^{[i]} = \begin{cases} \bar{P}_{pub}^{[i]} + a_i, & i \in S \\ \bar{P}_{pub}^{[i]} - b_i, & i \in S^c \end{cases} \quad (140)$$

where a_i are chosen such that $0 \leq a_i \leq 1 - \bar{P}_{pub}^{[i]}$ among $i \in S$ while satisfying $\sum_{i \in S} a_i = \alpha$, and b_i are chosen such that $0 \leq b_i \leq \bar{P}_{pub}^{[i]}$ among $i \in S^c$ while satisfying $\sum_{i \in S^c} b_i = \alpha$. Note that these selections are feasible since $\sum_{i \in S} a_i \leq 1 - \bar{P}_{pub}(S) \leq \sum_{i \in S} (1 - \bar{P}_{pub}^{[i]})$ as $|S| \geq 1$, and $\sum_{i \in S^c} b_i \leq 1 - \bar{P}_{pub}(S) = \bar{P}_{pub}(S^c) = \sum_{i \in S^c} \bar{P}_{pub}^{[i]}$.

With the $\hat{P}_{prv}$ in (140), for any $S \subset [d]$ satisfying $1 \leq |S| \leq d-1$, we have,

$$\hat{P}_{prv}(S) = \min\{\bar{P}_{pub}(S) + \gamma, 1\}. \quad (141)$$

Since the upper bound in (137) is satisfied with equality in (141),

$$\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \bar{P}_{prv}(S) = \min\{\bar{P}_{pub}(S) + \gamma, 1\}, \quad \forall S \subset [d], \quad 1 \leq |S| \leq d-1. \quad (142)$$

Plugging (142) back in (134) completes the proof of Lemma E.1. ■

F. Details of Remark 4.5

In this section, we provide the reasoning behind our approximation for the optimal noise distribution in Remark 4.5. The main optimization in Theorem 4.2 can be written as,

$$\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) = \min \left\{ \min_{P_{nse} \in \mathcal{A}} g(P_{nse}), \min_{P_{nse} \in \Delta_d \setminus \mathcal{A}} g(P_{nse}) \right\} \quad (143)$$

where $\mathcal{A} = \left\{ P_{nse} \in \Delta_d : \sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]} \geq \gamma \right\}$ and,

$$g(P_{nse}) = \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \quad (144)$$

for any given $P_{nse} \in \Delta_d$ with $\min_i P_{nse}^{[i]} = N_{min_1}$. Next, we consider the two cases $P_{nse} \in \mathcal{A}$ and $P_{nse} \in \mathcal{A}^c$ in (143) separately.

Lemma F.1. For given $\bar{P}_{pub} \in \Delta_d$ and $\gamma \in (0, 1)$,

$$\arg \min_{P_{nse} \in \mathcal{A}} g(P_{nse}) = \text{MIFR}(\bar{P}_{pub}, t_1) \quad (145)$$

where $t_1 = \arg \min_{t \in \mathcal{F}} \beta_{min}(t, t)(\gamma + \sum_{i: \bar{P}_{pub}^{[i]} \leq t} (t - \bar{P}_{pub}^{[i]}))$ with $\mathcal{F} = \{t : \sum_{i: \bar{P}_{pub}^{[i]} \leq \text{MIFR}^{[i]}(\bar{P}_{pub}, t)} \bar{P}_{pub}^{[i]} \geq \gamma\}$.

The proof of Lemma F.1 is given in Sec. F.1. Lemma F.1 characterizes the optimal solution to the first component in the minimization of (143). Next, we analyze the second component. Consider the set of distributions $\mathcal{A}^c = \Delta_d \setminus \mathcal{A}$.

$$\mathcal{A}^c = \{P_{nse} \in \Delta_d : \sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]} < \gamma\} \quad (146)$$

Lemma F.2. For given $\bar{P}_{pub} \in \Delta_d$ and $\gamma \in (0, 1)$, $\text{WF}(\bar{P}_{pub}, t_2) \in \mathcal{A}^c$, and

$$\min_{P_{nse} \in \mathcal{A}^c} g(P_{nse}) \leq g(\text{WF}(\bar{P}_{pub}, t_2)) \quad (147)$$

where $t_2 = \arg \min_{t \in \mathcal{G}} \beta_{min}(t, t)\Phi(t)$ with

$$\Phi(t) = \max_{k_t \leq u \leq d-1} \min \left\{ \gamma - \sum_{i=1}^u (\bar{P}_{pub}^{[i]} - \text{WF}^{[i]}(\bar{P}_{pub}, t)), \sum_{i=1}^u \text{WF}^{[i]}(\bar{P}_{pub}, t) \right\} \quad (148)$$

$k_t = \sum_{i=1}^d \mathbf{1}_{\{\bar{P}_{pub}^{[i]} \leq t\}}$, and $\mathcal{G} = \{t : \sum_{i: \bar{P}_{pub}^{[i]} \leq t} \bar{P}_{pub}^{[i]} < \gamma\}$.

The proof of Lemma F.2 is given in Sec. F.2. Combining Lemmas F.1 and F.2 with (143), we have,

$$\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \leq \min \{g(\text{MIFR}(\bar{P}_{pub}, t_1)), g(\text{WF}(\bar{P}_{pub}, t_2))\} \quad (149)$$

which corresponds to the suboptimal approximation of P_{nse}^* in Remark 4.5.

Fig. 7 shows the histogram of the differences between the objective values (minimax TV distance $g(P_{nse}) = \min_{\beta \geq \beta_{min}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn})$) obtained using the optimal noise distribution P_{nse}^* from Theorem 4.2 and the approximation $\hat{P}_{nse}^*$ from Remark 4.5. The histograms are computed over a grid of public distributions $\bar{P}_{pub}$, with $\gamma = 0.35$, $n = 100$, $m = 80$, $\varepsilon = 5$, and $\delta = 1/n$.

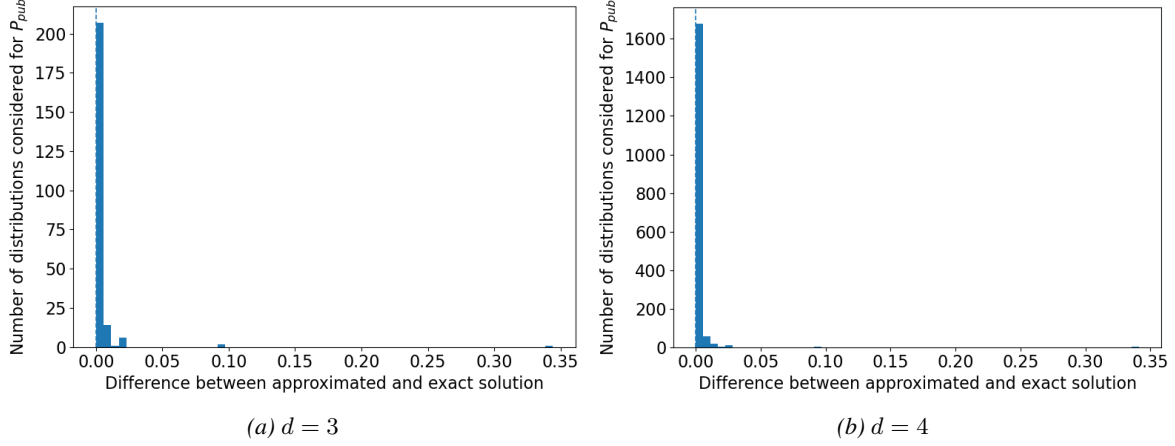


Figure 7. Histograms over candidate $\bar{P}_{pub}$ distributions, showing the difference between the minimax TV distance $g(P_{nse}) = \min_{\beta \geq \beta_{min}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma}, \bar{P}_{pub}} \text{TV}(\bar{P}_{prv}, P_{syn})$ considering the optimal P_{nse}^* from Theorem 4.2 and the approximation $\hat{P}_{nse}^*$ in Remark 4.5.

F.1. Proof of Lemma F.1

Proof: For a fixed $P_{nse} \in \mathcal{A}$ with $\min_i P_{nse}^{[i]} = N_{min_1}$,

$$g(P_{nse}) = \min_{\substack{\beta \in [0,1] \\ \beta \geq \beta_{min}(N_{min_1}, N_{min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma}, \bar{P}_{pub}} \text{TV}(\bar{P}_{prv}, P_{syn}) \quad (150)$$

$$= \beta_{min}(N_{min_1}, N_{min_1}) \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma}, \bar{P}_{pub}} \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (151)$$

Recall from Lemma E.1 that for any fixed P_{nse} :

$$\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma}, \bar{P}_{pub}} \text{TV}(\bar{P}_{prv}, P_{nse}) = \max_{\substack{S \subseteq [d] \\ 1 \leq |S| \leq d-1}} [\min \{ \bar{P}_{pub}(S) + \gamma, 1 \} - P_{nse}(S)] \quad (152)$$

Note that for any fixed $\bar{P}_{pub} \in \Delta_d$ and $P_{nse} \in \mathcal{A}$,

$$\text{TV}(P_{nse}, \bar{P}_{pub}) = P_{nse}(B) - \bar{P}_{pub}(B) \quad (153)$$

where $B = \{i : \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}\}$. Since we are considering $P_{nse} \in \mathcal{A}$, we have,

$$\bar{P}_{pub}(B) \geq \gamma \implies \bar{P}_{pub}(B^c) < 1 - \gamma \quad (154)$$

Thus, when $S = B^c$, we have,

$$\min \{ \bar{P}_{pub}(B^c) + \gamma, 1 \} - P_{nse}(B^c) = \bar{P}_{pub}(B^c) + \gamma - P_{nse}(B^c) = \text{TV}(\bar{P}_{pub}, P_{nse}) + \gamma \quad (155)$$

At the same time, we have:

$$\text{TV}(\bar{P}_{prv}, P_{nse}) \leq \text{TV}(\bar{P}_{prv}, \bar{P}_{pub}) + \text{TV}(\bar{P}_{pub}, P_{nse}) \leq \gamma + \text{TV}(\bar{P}_{pub}, P_{nse}) \quad (156)$$

Since the upper bound in (156) is satisfied with equality in (155) when $S = B^c$, combining (152), (155), and (156), we get:

$$\max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma}, \bar{P}_{pub}} \text{TV}(\bar{P}_{prv}, P_{nse}) = \gamma + \text{TV}(\bar{P}_{pub}, P_{nse}) \quad (157)$$

Continuing from (151), for any $P_{nse} \in \mathcal{A}$,

$$g(P_{nse}) = \beta_{min}(N_{min_1}, N_{min_1})(\gamma + \text{TV}(\bar{P}_{pub}, P_{nse})) \quad (158)$$

Consequently,

$$\min_{P_{nse} \in \mathcal{A}} g(P_{nse}) = \min_{0 \leq t \leq \frac{1}{d}} \beta_{min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A} \\ \min_i P_{nse}^{[i]} = t}} (\gamma + \text{TV}(\bar{P}_{pub}, P_{nse})) \quad (159)$$

Claim F.3. Consider a fixed $\bar{P}_{pub} \in \Delta_d$ and a fixed $t \in [0, 1/d]$. If $\text{MIFR}(\bar{P}_{pub}, t) \notin \mathcal{A}$, there exists no other P_{nse} with $\min_i P_{nse}^{[i]} = t$ satisfying $P_{nse} \in \mathcal{A}$.

Proof: To satisfy $\min_i P_{nse}^{[i]} = t$, we need $P_{nse}^{[i]} \geq t, \forall i$. Note that,

$$\arg \max_{\substack{P_{nse} \in \Delta_d \\ \min_i P_{nse}^{[i]} = t}} \sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]} = \text{MIFR}(\bar{P}_{pub}, t) \quad (160)$$

as $\text{MIFR}(\bar{P}_{pub}, t)$ increases the elements of $\bar{P}_{pub}$ below t exactly up to t , and removes the added mass from larger elements of $\bar{P}_{pub}$ such that the unchanged probability mass is maximized (see Def. 4.4). This is the exact definition of maximizing $\sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]}$ while satisfying the floor constraint. Thus, for a given $t \in [0, 1/d]$, if $\sum_{i: \bar{P}_{pub}^{[i]} \leq \text{MIFR}^{[i]}(\bar{P}_{pub}, t)} \bar{P}_{pub}^{[i]} < \gamma$, we have,

$$\sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]} \leq \sum_{i: \bar{P}_{pub}^{[i]} \leq \text{MIFR}^{[i]}(\bar{P}_{pub}, t)} \bar{P}_{pub}^{[i]} < \gamma, \quad \forall P_{nse} \in \Delta_d, \quad \min_i P_{nse}^{[i]} = t \quad (161)$$

Thus, no other P_{nse} with $\min_i P_{nse}^{[i]} = t$ satisfies $P_{nse} \in \mathcal{A}$ for that specific t . ■

For the given $\bar{P}_{pub}$, let $\mathcal{F} \subseteq [0, 1/d]$ be the feasible set of t , for which $\text{MIFR}(\bar{P}_{pub}, t) \in \mathcal{A}$. In other words, $\mathcal{F}$ is the set of t values for which the set $\mathcal{A} \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\}$ is non-empty. Then, we can rewrite (159) as,

$$\min_{P_{nse} \in \mathcal{A}} g(P_{nse}) = \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A} \\ \min_i P_{nse}^{[i]} = t}} (\gamma + \text{TV}(\bar{P}_{pub}, P_{nse})) \quad (162)$$

$$= \min_{t \in \mathcal{F}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A} \\ \min_i P_{nse}^{[i]} = t}} (\gamma + \text{TV}(\bar{P}_{pub}, P_{nse})) \quad (163)$$

$$\geq \min_{t \in \mathcal{F}} \beta_{\min}(t, t) (\gamma + \sum_{i: \bar{P}_{pub}^{[i]} \leq t} (t - \bar{P}_{pub}^{[i]})) \quad (164)$$

as the minimum $\text{TV}(\bar{P}_{pub}, P_{nse})$ with the $\min_i P_{nse}^{[i]} = t$ constraint is achieved when only the smaller elements of $\bar{P}_{pub}$ are increased to t and the corresponding additional mass is removed from larger $\bar{P}_{pub}^{[i]}$. At the same time, from the definition of MIFR in Def. 4.4, we have,

$$g(\text{MIFR}(\bar{P}_{pub}, t^*)) = \beta_{\min}(t^*, t^*) (\gamma + \sum_{i: \bar{P}_{pub}^{[i]} \leq t^*} (t^* - \bar{P}_{pub}^{[i]})) \quad (165)$$

where $t^* = \arg \min_{t \in \mathcal{F}} \beta_{\min}(t, t) (\gamma + \sum_{i: \bar{P}_{pub}^{[i]} \leq t} (t - \bar{P}_{pub}^{[i]}))$. This completes the proof of Lemma F.1 as $\text{MIFR}(\bar{P}_{pub}, t^*)$ achieves the lower bound in (164) with equality. ■

F.2. Proof of Lemma F.2

Proof: Consider the second component in the minimization of (143):

$$\min_{P_{nse} \in \mathcal{A}^c} g(P_{nse}) = \min_{P_{nse} \in \mathcal{A}^c} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{syn}) \quad (166)$$

$$= \min_{P_{nse} \in \mathcal{A}^c} \beta_{\min}(N_{\min_1}, N_{\min_1}) \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (167)$$

$$= \min_{0 \leq t \leq \frac{1}{d}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A}^c \\ \min_i P_{nse}^{[i]} = t}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (168)$$

Next, we characterize the set of t values for which $\mathcal{A}^c \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\} \neq \emptyset$.

Claim F.4. For any given $\bar{P}_{pub} \in \Delta_d$, fix $t \in [0, 1/d]$. If $\text{WF}(\bar{P}_{pub}, t) \notin \mathcal{A}^c$, then, $\mathcal{A}^c \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\} = \emptyset$.

Proof: For any P_{nse} with $\min_i P_{nse}^{[i]} = t$, we have,

$$\sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]} \geq \sum_{i: \bar{P}_{pub}^{[i]} \leq t} \bar{P}_{pub}^{[i]} = \sum_{i: \bar{P}_{pub}^{[i]} \leq \text{WF}(\bar{P}_{pub}, t)} \bar{P}_{pub}^{[i]} \quad (169)$$

from the construction of $\text{WF}(\bar{P}_{pub}, t)$ in Def. 4.3. Now, consider the case where $\text{WF}(\bar{P}_{pub}, t) \notin \mathcal{A}^c$, i.e., $\sum_{i: \bar{P}_{pub}^{[i]} \leq \text{WF}(\bar{P}_{pub}, t)} \bar{P}_{pub}^{[i]} > \gamma$. Then, for any $P_{nse} \in \Delta_d$ satisfying $\min_i P_{nse}^{[i]} = t$, we have,

$$\gamma < \sum_{i: \bar{P}_{pub}^{[i]} \leq \text{WF}(\bar{P}_{pub}, t)} \bar{P}_{pub}^{[i]} = \sum_{i: \bar{P}_{pub}^{[i]} \leq t} \bar{P}_{pub}^{[i]} \leq \sum_{i: \bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}} \bar{P}_{pub}^{[i]} \implies \mathcal{A}^c \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\} = \emptyset \quad (170)$$

■

Claim F.4 states that $\mathcal{A}^c \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\} \neq \emptyset$ only for $t \in \mathcal{G}$ where $\mathcal{G} = \{t : \sum_{i: \bar{P}_{pub}^{[i]} \leq t} \bar{P}_{pub}^{[i]} < \gamma\}$. Now, we continue from (168):

$$\min_{P_{nse} \in \mathcal{A}^c} g(P_{nse}) = \min_{t \in \mathcal{G}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A}^c \\ \min_i P_{nse}^{[i]} = t}} \max_{\bar{P}_{prv} \in \mathcal{B}_{\gamma, \bar{P}_{pub}}} \text{TV}(\bar{P}_{prv}, P_{nse}) \quad (171)$$

$$= \min_{t \in \mathcal{G}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A}^c \\ \min_i P_{nse}^{[i]} = t}} \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} [\min \{\bar{P}_{pub}(S) + \gamma, 1\} - P_{nse}(S)] \quad (172)$$

$$= \min_{t \in \mathcal{G}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A}^c \\ \min_i P_{nse}^{[i]} = t}} \max_{\substack{S \subset [d] \\ 1 \leq |S| \leq d-1}} \min \{\bar{P}_{pub}(S) - P_{nse}(S) + \gamma, 1 - P_{nse}(S)\} \quad (173)$$

$$= \min_{t \in \mathcal{G}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A}^c \\ \min_i P_{nse}^{[i]} = t}} \max_{\substack{S^c \subset [d] \\ 1 \leq |S^c| \leq d-1}} \min \{P_{nse}(S^c) - \bar{P}_{pub}(S^c) + \gamma, P_{nse}(S^c)\} \quad (174)$$

$$= \min_{t \in \mathcal{G}} \beta_{\min}(t, t) \min_{\substack{P_{nse} \in \mathcal{A}^c \\ \min_i P_{nse}^{[i]} = t}} \max_{\substack{R \subset [d] \\ 1 \leq |R| \leq d-1}} \min \{P_{nse}(R) - \bar{P}_{pub}(R) + \gamma, P_{nse}(R)\} \quad (175)$$

where (173) follows from Lemma E.1, and $R = S^c$ is used for the convenience in notation.

Claim F.5. For a given P_{nse} , let $L = \{i : P_{nse}^{[i]} \geq \bar{P}_{pub}^{[i]}\}$. Then, for any $R \subset [d]$, $1 \leq |R| \leq d-1$,

$$\min \{P_{nse}(R) - \bar{P}_{pub}(R) + \gamma, P_{nse}(R)\} \leq \min \{P_{nse}(R \cup L) - \bar{P}_{pub}(R \cup L) + \gamma, P_{nse}(R \cup L)\} \quad (176)$$

Proof: (176) holds since each $i \in L$ contributes to $P_{nse}^{[i]} - \bar{P}_{pub}^{[i]} \geq 0$ and $P_{nse}^{[i]} \geq 0$. ■

Assume that the indices are arranged such that $\bar{P}_{pub}^{[i]} - P_{nse}^{[i]} \leq \bar{P}_{pub}^{[i+1]} - P_{nse}^{[i+1]}$, $\forall i$, and let $k(P_{nse}) = \sum_{i=1}^d \mathbf{1}_{\{\bar{P}_{pub}^{[i]} \leq P_{nse}^{[i]}\}}$. Recall that $L = \{1, \dots, k(P_{nse})\}$ for a given P_{nse} . Then,

$$\begin{aligned} & \max_{\substack{R \subset [d] \\ 1 \leq |R| \leq d-1}} \min \{P_{nse}(R) - \bar{P}_{pub}(R) + \gamma, P_{nse}(R)\} \\ &= \max_{\substack{C \subset \{k(P_{nse})+1, \dots, d\} \\ 0 \leq |C| \leq d-1-k(P_{nse})}} \min \{P_{nse}(C \cup L) - \bar{P}_{pub}(C \cup L) + \gamma, P_{nse}(C \cup L)\} \end{aligned} \quad (177)$$

$$\geq \max_{k(P_{nse}) \leq u \leq d-1} \min \left\{ \gamma - \sum_{i=1}^u (\bar{P}_{pub}^{[i]} - P_{nse}^{[i]}), \sum_{i=1}^u P_{nse}^{[i]} \right\} \quad (178)$$

since only the ordered subsets C are considered in (178). Note that $\text{WF}(\bar{P}_{pub}, t) \in \mathcal{A}^c \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\}$ for any feasible t . Then,

$$k(\text{WF}(\bar{P}_{pub}, t)) = \sum_{i=1}^d \mathbf{1}_{\{\bar{P}_{pub}^{[i]} \leq \text{WF}^{[i]}(\bar{P}_{pub}, t)\}} = \sum_{i=1}^d \mathbf{1}_{\{\bar{P}_{pub}^{[i]} \leq t\}} = k_t \quad (179)$$

Define $\Phi(t) = \max_{k_t \leq u \leq d-1} \min \left\{ \gamma - \sum_{i=1}^u (\bar{P}_{pub}^{[i]} - \text{WF}^{[i]}(\bar{P}_{pub}, t)), \sum_{i=1}^u \text{WF}^{[i]}(\bar{P}_{pub}, t) \right\}$, and compute:

$$t_2 = \arg \max_{t \in \mathcal{G}} \beta_{\min}(t, t) \Phi(t) \quad (180)$$

Since $\text{WF}(\bar{P}_{pub}, t) \in \mathcal{A}^c \cap \{P_{nse} : \min_i P_{nse}^{[i]} = t\}$, $\forall t \in \mathcal{G}$ and $t_2 \in \mathcal{G}$, we have

$$\min_{P_{nse} \in \mathcal{A}^c} g(P_{nse}) \leq g(\text{WF}(\bar{P}_{pub}, t_2)) \quad (181)$$

■

G. Proof of Corollary 4.6

Corollary 4.6 restated: For any D_{prv} , $m \geq 1$, $\varepsilon > 0$ and $\delta \in (0, 1)$, the asymptotic solution to (3) with $B_{\gamma, S} = \Delta_d$ is given by,

$$\min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{P_{prv} \in \Delta_d} \text{TV}(\bar{P}_{prv}, P_{syn}) = \left(1 - \frac{1}{d}\right) \beta_{\min}(1/d, 1/d) \quad (182)$$

and the corresponding optimum β^* and P_{nse}^* are given by,

$$P_{nse}^* = \frac{1}{d} \mathbf{1}_d \quad (183)$$

$$\beta^* = \beta_{\min}(1/d, 1/d) \quad (184)$$

Proof. Let $\bar{P}_{prv}^{[i]}$ and $P_{nse}^{[i]}$ denote the i th elements of $\bar{P}_{prv}$ and P_{nse} , respectively. Consider,

$$\begin{aligned} & \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \Delta_d} \text{TV}(\bar{P}_{prv}, (1 - \beta)\bar{P}_{prv} + \beta P_{nse}) \\ &= \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \max_{\bar{P}_{prv} \in \Delta_d} \frac{\beta}{2} \sum_{i=1}^d |\bar{P}_{prv}^{[i]} - P_{nse}^{[i]}| \end{aligned} \quad (185)$$

$$= \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_1})}} \frac{\beta}{2} \sum_{i=1, i \neq \arg \min_j P_{nse}^{[j]}}^d P_{nse}^{[i]} + \frac{\beta}{2} (1 - N_{\min_1}), \quad (\text{since } \min_j P_{nse}^{[j]} = N_{\min_1}) \quad (186)$$

$$= \min_{P_{nse} \in \Delta_d} \min_{\substack{\beta \in [0, 1] \\ \beta \geq \beta_{\min}(N_{\min_1}, N_{\min_2})}} \beta (1 - N_{\min_1}) \quad (187)$$

$$= \min_{P_{nse} \in \Delta_d} \beta_{\min}(N_{\min_1}, N_{\min_1}) (1 - N_{\min_1}) \quad (188)$$

From Lemma C.5, we know that $\beta_{\min}(N_{\min_1}, N_{\min_1})$ is decreasing in $N_{\min_1}$. Then, as $\beta_{\min}(N_{\min_1}, N_{\min_1})$ and $1 - N_{\min_1}$ are both decreasing in $N_{\min_1}$, we have the optimal $N_{\min_1} = \frac{1}{d}$, and the corresponding optimal $P_{nse}^* = \frac{1}{d} \mathbf{1}_d$. The resulting minimum TV is given in (182). $\square$

H. Extension to General Discrete Distributions

In this section, we provide the extension of the privacy constraint when $\bar{P}_{prv}$ and $\bar{P}_{pub}$ are arbitrary discrete distributions (not only normalized histograms). The privacy constraint imposes a lower bound on β for a given P_{nse} (see Lemma H.1). The same approach in Theorem 4.2 and Remark 4.5 is then taken to obtain the optimum noise distributions.

Lemma H.1. Fix $\varepsilon > 0$, $\delta \in (0, 1)$, and the required number of synthetic samples $m \geq 1$. Let $P_{\text{nse}} \in \Delta_d$ be any noise distribution and define $p := \min_{y \in \{1, \dots, d\}} P_{\text{nse}}(y) > 0$. Define the sensitivity as $s = \max_i |\bar{P}_{\text{prv}}^{[i]} - \bar{P}'_{\text{prv}}^{[i]}|$, considering all neighboring private distributions $\bar{P}_{\text{prv}}, \bar{P}'_{\text{prv}}$ corresponding to neighboring datasets $D_{\text{prv}}, D'_{\text{prv}}$. Then, for every $\beta \geq \beta_{\min}$, the m synthetic samples drawn from P_{syn} in (1) satisfy the DP constraint in (2).

Proof. Consider the privacy loss random variable (PLRV) in (2):

$$\log \frac{\mathbb{P}(Y_{P_{\text{syn}}} = \{y_1, \dots, y_m\})}{\mathbb{P}(Y_{P'_{\text{syn}}} = \{y_1, \dots, y_m\})} = \sum_{i=1}^m \log \frac{P_{\text{syn}}(y_i)}{P'_{\text{syn}}(y_i)} \quad (189)$$

Define $L(y_i) = \log \frac{P_{\text{syn}}(y_i)}{P'_{\text{syn}}(y_i)}$, which are i.i.d. Note that,

$$P_{\text{syn}}(y) = (1 - \beta)\bar{P}_{\text{prv}}(y) + \beta P_{\text{nse}}(y) \quad (190)$$

$$P'_{\text{syn}}(y) = (1 - \beta)\bar{P}'_{\text{prv}}(y) + \beta P_{\text{nse}}(y) \quad (191)$$

where,

$$\bar{P}'_{\text{prv}}(y) - \min\{s, \bar{P}'_{\text{prv}}(y)\} \leq \bar{P}_{\text{prv}}(y) \leq \bar{P}'_{\text{prv}}(y) + \min\{s, 1 - \bar{P}'_{\text{prv}}(y)\}, \quad \forall y \in \{1, \dots, d\} \quad (192)$$

such that $\sum_{y=1}^d \bar{P}_{\text{prv}}(y) = 1$. Then,

$$1 - \frac{(1 - \beta) \min\{s, \bar{P}'_{\text{prv}}(y)\}}{P'_{\text{syn}}(y)} \leq \frac{P_{\text{syn}}(y)}{P'_{\text{syn}}(y)} \leq 1 + \frac{(1 - \beta) \min\{s, 1 - \bar{P}'_{\text{prv}}(y)\}}{P'_{\text{syn}}(y)} \quad (193)$$

Note that if $s \leq \bar{P}'_{\text{prv}}(y)$ the lower bound simplifies to:

$$1 - \frac{(1 - \beta)s}{P'_{\text{syn}}(y)} \geq 1 - \frac{(1 - \beta)s}{(1 - \beta)s + \beta p} = 1 - \frac{1}{1 + \frac{\beta p}{(1 - \beta)s}} \quad (194)$$

If $s > \bar{P}'_{\text{prv}}(y)$ the lower bound simplifies to:

$$1 - \frac{(1 - \beta)\bar{P}'_{\text{prv}}(y)}{(1 - \beta)\bar{P}'_{\text{prv}}(y) + \beta p} \geq 1 - \frac{1}{1 + \frac{\beta p}{(1 - \beta)s}} \quad (195)$$

Similarly, the upper bound in (193) is simplified to $1 + \frac{(1 - \beta)s}{\beta p}$. Thus, we have,

$$L = 1 - \frac{1}{1 + \frac{\beta p}{(1 - \beta)s}} \leq \frac{P_{\text{syn}}(y)}{P'_{\text{syn}}(y)} \leq 1 + \frac{(1 - \beta)s}{\beta p} = U \quad (196)$$

Therefore, any $t = \frac{P_{\text{syn}}(y)}{P'_{\text{syn}}(y)}$ can be written as $t = \theta L + (1 - \theta)U$ for $\theta = \frac{U - t}{U - L} \in [0, 1]$.

To satisfy (2), we need:

$$\mathbb{P}\left(\sum_{i=1}^m \log \frac{P_{\text{syn}}(y_i)}{P'_{\text{syn}}(y_i)} > \varepsilon\right) \leq \delta \implies \mathbb{P}\left(e^{\lambda \sum_{i=1}^m \log \frac{P_{\text{syn}}(y_i)}{P'_{\text{syn}}(y_i)}} > e^{\lambda \varepsilon}\right) \leq e^{-\lambda \varepsilon} \mathbb{E}\left[e^{\lambda \sum_{i=1}^m \log \frac{P_{\text{syn}}(y_i)}{P'_{\text{syn}}(y_i)}}\right] \leq \delta \quad (197)$$

Now, let $\lambda = \alpha - 1$ and consider,

$$\mathbb{E} \left[e^{\lambda \sum_{i=1}^m \log \frac{P_{syn}(y_i)}{P'_{syn}(y_i)}} \right] = \sum_{y_1, \dots, y_m} \prod_{i=1}^m P_{syn}(y_i) e^{(\alpha-1) \sum_{i=1}^m \log \frac{P_{syn}(y_i)}{P'_{syn}(y_i)}} \quad (198)$$

$$= \sum_{y_1, \dots, y_m} \prod_{i=1}^m P_{syn}(y_i) \left(\prod_{i=1}^m \frac{P_{syn}(y_i)}{P'_{syn}(y_i)} \right)^{\alpha-1} \quad (199)$$

$$= \prod_{i=1}^m \sum_{y_i} P_{syn}(y_i)^\alpha P'_{syn}(y_i)^{1-\alpha} \quad (200)$$

$$= \prod_{i=1}^m e^{\log \sum_{y_i} P_{syn}(y_i)^\alpha P'_{syn}(y_i)^{1-\alpha}} \quad (201)$$

$$= \prod_{i=1}^m e^{(\alpha-1) D_\alpha(P_{syn} || P'_{syn})} \quad (202)$$

$$= e^{(\alpha-1) m D_\alpha(P_{syn} || P'_{syn})} \quad (203)$$

From (197), we have the privacy constraint simplified to,

$$m \log \left(\sum_y P'_{syn}(y) \left(\frac{P_{syn}(y)}{P'_{syn}(y)} \right)^\alpha \right) - (\alpha - 1)\varepsilon \leq \log \delta \quad (204)$$

Note that $\mathbb{E}_{P'_{syn}}[t^\alpha] = \sum_y P'_{syn}(y) \left(\frac{P_{syn}(y)}{P'_{syn}(y)} \right)^\alpha$ and $h(t) = t^\alpha$ is convex. Therefore,

$$t^\alpha = (\theta L + (1 - \theta)U)^\alpha \leq \theta L^\alpha + (1 - \theta)U^\alpha \quad (205)$$

$$\implies \mathbb{E}_{P'_{syn}}[t^\alpha] \leq \mathbb{E}_{P'_{syn}}[\theta] L^\alpha + \mathbb{E}_{P'_{syn}}[1 - \theta] U^\alpha \quad (206)$$

$$= \frac{U - 1}{U - L} L^\alpha + \frac{1 - L}{U - L} U^\alpha \quad (207)$$

$$= \frac{\eta + 1}{\eta + 2} \frac{1}{(\eta + 1)^\alpha} + \frac{1}{\eta + 2} (1 + \eta)^\alpha \quad (208)$$

where $\eta = \frac{(1-\beta)s}{\beta p}$. Then, from (204), the privacy constraint simplifies to:

$$m \log \left(\frac{\eta + 1}{\eta + 2} \frac{1}{(\eta + 1)^\alpha} + \frac{1}{\eta + 2} (1 + \eta)^\alpha \right) - (\alpha - 1)\varepsilon \leq \log \delta \quad (209)$$

Next, we find the optimum $\alpha^* > 1$ that minimizes the LHS of (209) with other parameters fixed. Let $f(\alpha) = m \log \left(\frac{\eta+1}{\eta+2} \frac{1}{(\eta+1)^\alpha} + \frac{1}{\eta+2} (1+\eta)^\alpha \right) - (\alpha - 1)\varepsilon$ and $h(\alpha) = \frac{\eta+1}{\eta+2} \frac{1}{(\eta+1)^\alpha} + \frac{1}{\eta+2} (1+\eta)^\alpha$. Then,

$$f'(\alpha) = m \frac{h'(\alpha)}{h(\alpha)} - \varepsilon \quad (210)$$

$$h'(\alpha) = \frac{(\eta+1)^\alpha}{\eta+2} \log(\eta+1) - \frac{\eta+1}{\eta+2} (\eta+1)^{-\alpha} \log(\eta+1) = \frac{\log(\eta+1)}{\eta+2} ((\eta+1)^\alpha - (\eta+1)^{1-\alpha}) \quad (211)$$

$$h''(\alpha) = \frac{\log^2(\eta+1)}{\eta+2} ((\eta+1)^\alpha + (\eta+1)^{1-\alpha}) \quad (212)$$

$$f''(\alpha) = \frac{2m(h''(\alpha)h(\alpha) - h'(\alpha)^2)}{h^2(\alpha)} = \frac{m}{h^2(\alpha)} \frac{\log^2(\eta+1)}{(\eta+2)^2} (\eta+1) > 0 \quad (213)$$

Thus, $f(\alpha)$ has a unique minimum at $m \frac{h'(\alpha)}{h(\alpha)} - \varepsilon = 0$, which simplifies to:

$$\alpha^* = \frac{\log \left(\frac{(\eta+1)(m \log(\eta+1) + \varepsilon)}{m \log(\eta+1) - \varepsilon} \right)}{2 \log(\eta+1)} \quad (214)$$

which requires $\varepsilon < m \log(\eta + 1)$. For $\varepsilon \geq m \log(\eta + 1)$, we can easily show that $f'(\alpha) < 0$ and the optimum α is given by $\alpha^* \rightarrow \infty$, at which $f(\alpha) = -\infty$, satisfying (204) for any β .

Note that (214) needs to satisfy $\alpha^* > 1$. For this, we need:

$$\frac{(1-\beta)s}{\beta p} \left(\log \left(1 + \frac{(1-\beta)s}{\beta p} \right) - \frac{\varepsilon}{m} \right) \leq \frac{2\varepsilon}{m} \quad (215)$$

Since the LHS of (215) is decreasing in β , any $\beta \geq \tilde{\beta}_1$ satisfies (215), where $\tilde{\beta}_1$ is the solution to β in (215) with equality.

Now, plugging α^* back in the privacy constraint in (209) gives:

$$m \log \left(\frac{2m \log(\eta + 1)}{\eta + 2} \sqrt{\frac{\eta + 1}{m^2 \log^2(\eta + 1) - \varepsilon^2}} \right) - \frac{\varepsilon}{2 \log(\eta + 1)} \left(\log \left(\frac{m \log(\eta + 1) + \varepsilon}{m \log(\eta + 1) - \varepsilon} \right) - \log(\eta + 1) \right) \leq \log \delta \quad (216)$$

with $\eta = \frac{(1-\beta)s}{\beta p}$. Since the LHS of (216) is decreasing in β , any $\beta \geq \tilde{\beta}_2$ satisfies (216), where $\tilde{\beta}_2$ is the solution to β in (216) with equality. Thus, we have,

$$\beta_{min} = \max\{\tilde{\beta}_1, \tilde{\beta}_2\} \quad (217)$$

where $\tilde{\beta}_1, \tilde{\beta}_2$ are functions of $\varepsilon, \delta, m, n, s$, and $p = \min_{y \in \{1, \dots, d\}} P_{nse}(y)$.

□

I. Implementation Details

I.1. Implementation Details for the Experiment on Person Activity

We evaluate tabular DP synthetic data generation on the Person Activity dataset under a subject-based split to ensure there is no user overlap between private and public data. The dataset contains 164,860 total samples with 5 numerical features and 1 categorical feature, and the prediction target consists of 11 activity classes. We construct the private dataset from users with IDs $\{1, 2, 4\}$ (107,237 samples) and the public dataset from users with IDs $\{3, 5\}$ (57,623 samples).

Across methods, we consider privacy budgets $\varepsilon \in \{1.0, 2.0, 4.0\}$. We set $\delta = 1/(n \log n)$, where n is the private dataset size. For the PE-based synthesis pipeline, we use 15 epochs with 3 sampling epochs, generate 5000 samples per sampling stage, and do 3 variations for candidate generation. For the PE+PubMIX variant, we keep the same base configuration and additionally set γ with 0.9. For graphical-model baselines AIM, GSD, and GEM, we use the default configuration with degree=2. For PrivSyn, we use threshold with 20000 and 50 iterations as the default setting.

I.2. Implementation Details for the Experiment on Yelp

For text synthesis, we use the Yelp reviews corpus as the private training set (approximately 1.9M samples) with the standard dev/test splits. As the generator LLM, we use Llama-2-7B (Touvron et al., 2023) and enable stochastic decoding. Using PE, we run the synthesis procedure for 10 epochs. At each epoch, we generate 5,000 synthetic samples and set the variation degree to 0.5. To support selection via semantic similarity, we embed sentences using Sentence-BERT (Reimers & Gurevych, 2019) and perform nearest-neighbor matching with L2 distance. For privacy accounting, we consider $\varepsilon \in \{1, 2, 4\}$ and use the corresponding noise multipliers $\{15.34, 8.03, 4.24\}$, respectively. In addition to PE-style synthesis, we include PE+PubMix configured with $\delta = 10^{-5}$ and $\gamma = 0.1$.

For downstream utility evaluation, we fine-tune a RoBERTa-based classifier/regressor (Liu et al., 2019) on the generated synthetic reviews for two tasks: (i) business category prediction (10-class classification) and (ii) review star prediction (regression over 1–5). We train for 3 epochs with batch size 64, learning rate 3×10^{-5} , and maximum sequence length 64. We select the best checkpoint based on validation performance. We report accuracy for the category task and MAE for the star regression task.

The results for the same experiment as in Table 2, with a smaller public dataset (reviews from only MA as opposed to 10 states) is given in Table 3.

Table 3. Utility on Yelp Reviews under varying privacy budgets (public dataset = Google reviews from MA). We report mean $\pm$ std over multiple runs. Category is classification accuracy (% , higher is better) and Rating is RMSE (lower is better).

Metric	Method	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 4$
Category ($\uparrow$)	PE (Xie et al., 2024)	71.67 $\pm$ 0.21	70.69 $\pm$ 0.41	69.93 $\pm$ 0.38
	PUBMIX	71.03 $\pm$ 0.57	72.02 $\pm$ 0.42	71.63 $\pm$ 0.06
Rating ($\downarrow$)	PE (Xie et al., 2024)	0.898 $\pm$ 0.025	0.905 $\pm$ 0.009	0.889 $\pm$ 0.021
	PUBMIX	0.874 $\pm$ 0.011	0.887 $\pm$ 0.027	0.869 $\pm$ 0.010